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Research paper

Health-FedNet: A privacy-preserving federated learning framework for scalable and secure healthcare analytics

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A R T I C L E I N F O

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A B S T R A C T

The growing demand for privacy-preserving healthcare analytics necessitates solutions that comply with global regulatory standards such as HIPAA and GDPR while maintaining high diagnostic accuracy. This paper introduces Health-FedNet, a federated learning (FL) framework designed for secure, decentralized model training across multiple healthcare institutions without transferring raw patient data. Health-FedNet integrates Differential Privacy (DP), Homomorphic Encryption (HE), and an Adaptive Node Weighting Mechanism to enhance privacy, scalability, and robustness against heterogeneous data distributions. Evaluated on the MIMIC-III clinical database, Health-FedNet achieved a 12 % increase in diagnostic accuracy compared to centralized models, with statistical significance confirmed through a Wilcoxon signed-rank test ($p < 0.01$). The adaptive weighting mechanism prioritizes high-quality data sources, ensuring robust learning and model convergence. Furthermore, Health-FedNet supports real-time streaming updates, optimizing communication overhead and latency, making it suitable for time-sensitive clinical scenarios. Encryption mechanisms are designed to maintain privacy during transmission, aligning with international data protection standards. Future work will explore cross-border scalability, multi-institutional real-time synchronization, and edge-based streaming analytics to further enhance global healthcare collaborations. Health-FedNet represents a scalable, privacy-focused solution for modern healthcare analytics, advancing secure and efficient federated learning in clinical settings.

1. Introduction

Artificial Intelligence (AI) is revolutionizing the healthcare industry by enabling advanced predictive modeling, disease diagnostics, and personalized treatment plans [1]. Through machine learning (ML) techniques, healthcare providers can leverage large-scale patient data to generate actionable insights, improve clinical outcomes, and increase operational efficiencies [2]. AI has found significant applications across diverse domains, including chronic disease prediction, medical imaging, drug discovery, and remote patient monitoring [3]. Despite its vast potential, the effectiveness of AI in healthcare largely depends on the availability of comprehensive, high-quality datasets that include sensitive patient information, such as medical history, genetic data, and diagnostic records [4]. This reliance on sensitive data brings forth serious concerns regarding data privacy, security, and regulatory compliance [5].

Patient data is traditionally stored in decentralized silos spread across hospitals, clinics, and research labs, making the aggregation of datasets for robust predictive modeling challenging [6]. Centralized approaches that consolidate sensitive patient data into a single repository pose substantial privacy risks, as evidenced by high-profile data breaches such as the Anthem data breach, which compromised the records of >79 million individuals [7]. Such data breaches and strict privacy laws necessitate decentralized approaches like FL, which preserve data locality and support regulatory compliance. Moreover, stringent privacy regulations like the Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) in Europe further restrict data sharing, creating a critical need for innovative solutions that balance data utility and patient confidentiality [8].

1.1. Federated learning for decentralized healthcare analytics

As a privacy-preserving machine learning paradigm in decentralized environments, Federated Learning (FL) has emerged as a viable solution. FL enables collaborative training of models across multiple institutions while maintaining local data privacy, significantly reducing the risks associated with centralized data aggregation [9]. Unlike traditional

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centralized methods, FL eliminates the need for data transfer, allowing individual nodes (e.g., hospitals, clinics) to train local models on-site. This decentralized approach not only preserves privacy but also complies with regulatory frameworks like HIPAA and GDPR, making it particularly suitable for healthcare applications.

However, existing FL frameworks exhibit critical vulnerabilities. Shared gradients from local updates can leak sensitive information, enabling adversaries to infer private data through gradient inversion techniques [10]. Furthermore, current FL models often struggle to scale efficiently with large healthcare datasets and maintain robustness across non-IID (non-independent and identically distributed) data distributions [11]. Techniques like differential privacy and homomorphic encryption have been explored to enhance security, but they typically introduce computational inefficiencies and degrade model accuracy, limiting their application in real-world healthcare settings [12].

Healthcare data is inherently fragmented across multiple institutions due to patient mobility and the decentralized nature of healthcare delivery. Traditional privacy-preserving methods like Secure Multi-Party Computation (SMPC) and centralized Differential Privacy (DP) are limited in scalability and real-time adaptability. SMPC involves high computational complexity and synchronization overhead, making it impractical for large-scale, real-time analytics across multiple hospitals. Centralized DP, while effective in anonymizing datasets, still requires raw data to be collected in a central server, violating HIPAA and GDPR compliance for medical records. In contrast, Federated Learning (FL) maintains data locality, allowing institutions to collaboratively train a global model without data movement, thus preserving privacy and institutional autonomy. Health-FedNet capitalizes on these strengths by combining FL with advanced encryption mechanisms, enabling secure, real-time learning across heterogeneous healthcare providers while maintaining regulatory compliance.

1.2. Motivation for health-FedNet

To address these challenges, this study introduces Health-FedNet, a federated learning framework designed to optimize data privacy, scalability, and model robustness in decentralized healthcare analytics. Health-FedNet is illustrated in Fig. 1, showcasing a system of decentralized data owners (A, B, C) who collaboratively train local models using their private datasets. Local model updates are encrypted with homomorphic encryption to ensure privacy and then sent securely to a trusted aggregator. The aggregator performs secure aggregation of these encrypted updates, maintaining compliance with HIPAA and GDPR standards. The global model is subsequently updated and shared with all nodes, ensuring data privacy and regulatory adherence.

The primary motivation of this research is to bridge the gap between data utility and privacy preservation in AI-driven healthcare applications. Health-FedNet addresses the limitations of existing FL frameworks by integrating differential privacy for local obfuscation [17], homomorphic encryption for secure parameter exchange [19], and an adaptive node weighting mechanism [20] to prioritize nodes based on data quality and reliability. This mechanism works like a classroom setting where students with clearer answers are given more credit—similarly, institutions with higher-quality models contribute more to the final global model. These design choices build upon foundational advances in privacy-preserving machine learning and intelligent system optimization [18], delivering a unified and regulation-compliant solution that enables healthcare institutions to collaboratively analyze sensitive data while preserving confidentiality and fostering trust.

1.3. Problem statement

The primary challenge in healthcare analytics is that data is fragmented across hospitals and institutions, which complicates effective data sharing while ensuring privacy and regulatory compliance [12–14]. Traditional centralized approaches are prone to data breaches and fail to comply with privacy regulations like HIPAA and GDPR. While FL offers a decentralized solution, existing FL frameworks fall short in guaranteeing privacy, scalability, and robustness to heterogeneous data [15]. Shared gradients in federated models can reveal sensitive information to adversaries, and existing encryption methods often suffer from low computational efficiency [16].

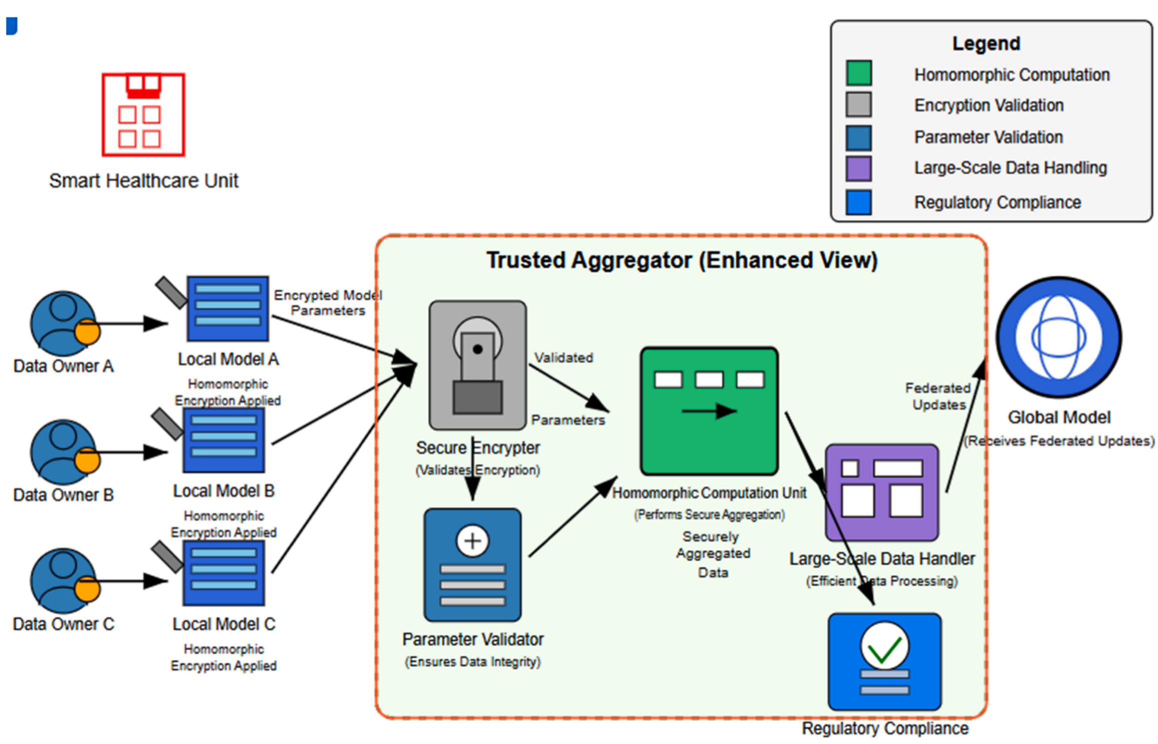

Fig. 1. Trusted Aggregator In Health-FedNet Architecture.

In response to these challenges, this paper presents Health-FedNet, a federated learning framework built on advanced privacy-preserving methods and adaptive mechanisms to ensure adequate privacy, compliance, and model accuracy in healthcare data analysis [17]. FL presents a decentralized solution but struggles to overcome privacy vulnerabilities and inconsistent data quality among institutions [18]. To address these gaps, Health-FedNet integrates differential privacy to obfuscate individual data contributions through noise addition, and homomorphic encryption to maintain data confidentiality during aggregation [19].

1.4. Contributions of health-FedNet

The contributions of Health-FedNet include:

1. Differential Privacy and Homomorphic Encryption: Health-FedNet combines differential privacy for local model updates and homomorphic encryption for secure parameter aggregation, ensuring data confidentiality and regulatory compliance [19].
2. Adaptive Node Weighting Mechanism: The adaptive node weighting mechanism prioritizes high-quality datasets, enabling more effective contributions to the global model, which enhances robustness and reduces bias [20].
3. Multi-Pronged Privacy Approach: By combining encryption and adaptive mechanisms, Health-FedNet guarantees compliance with HIPAA and GDPR, fostering trust and collaboration between institutions [20].
4. Scalable Decentralized Learning: Health-FedNet addresses scalability challenges through quantized encryption, optimizing memory usage and bandwidth efficiency [21].

1.5. Research objectives

The primary objectives of this study are as follows:

- To develop a privacy-preserving federated learning framework that ensures secure model training while complying with privacy regulations like HIPAA and GDPR [12,13].
- To implement an adaptive node weighting mechanism that prioritizes high-quality data sources, enhancing model robustness and diagnostic accuracy [20].
- To evaluate the proposed framework using the MIMIC-III dataset for chronic disease prediction, comparing its performance with centralized and existing federated models [21].
- To analyze the scalability, computational efficiency, and privacy guarantees of Health-FedNet under heterogeneous and large-scale healthcare data environments [22].

In Table 1, Madhavi et al. (2023), the effectiveness of adaptive privacy preserving methods in healthcare was shown, where adaptive methods can maintain the robustness of the model while complying with stringent privacy laws [17]. The key to being able to achieve collaborative analytics without compromising data confidentiality was this approach. FL is shown by Mehendale (2021) to have the potential to transform data sharing in hospitals and research institutions. It is the ability of federated systems to keep the data local that makes it less risky to store the data in the centralized manner and to keep it secure. Moon and Lee (2023) looked into the challenges in deploying FL in healthcare, such as scalability and communication overhead. To solve these challenges, they proposed privacy preserving techniques to make decentralized systems operate efficiently without exposing sensitive data [19]. In line with this, Najib et al. (2024) explored the unexploited potential of FL in achieving the balance between privacy assurance and analytical performance. However, their work identified critical gaps in existing methodologies, which often do not achieve high model accuracy while still providing adequate privacy [20]. Like Narmadha and Varalakshmi

Table 1
Comparative Table of previous Studies.

Attributes	Madhavi et al.	Patel et al.	Moon and Lee	Yang et al.
Techniques	Adaptive node weighting, differential privacy	Differential privacy, secure aggregation	Secure communication protocols, differential privacy	Homomorphic encryption, secure aggregation
Methodology	Applied adaptive methods prioritizing high-quality data nodes.	Developed the LEAF framework integrating privacy-preserving techniques.	Designed secure FL models with reduced communication overhead.	Integrated privacy techniques for secure medical data sharing.
Key Contributions	Enhanced privacy compliance and model robustness.	Enabled scalable FL with advanced privacy techniques.	Addressed communication and scalability challenges.	Demonstrated secure FL in pharmaceutical collaboration.
Limitations	Scalability issues and limited validation.	High communication overhead.	Restricted scalability and lack of real-world validation.	High computational complexity and limited dataset testing.
Applications	Decentralized diagnostic systems and regulatory compliance.	Multi-institutional healthcare collaboration.	Privacy-preserving patient monitoring systems.	Secure drug development and clinical trial management.

(2022) also, FL was used by Narmadha and Varalakshmi (2022) to explore the role of FL in regulatory compliance and how it can navigate the complex legal frameworks while facilitating cross-institutional data collaboration [21]. Innovative AI/ML architectures for addressing computational and privacy concerns in healthcare were introduced by Padmanaban (2024). Finally, these architectures demonstrated how tailored solutions could harmonize efficiency with security in data driven healthcare systems [22]. In a recent work, Patel et al. (2024) proposed the LEAF framework which specifically tackled privacy and scalability problems in federated healthcare ecosystems. Incorporating advanced privacy techniques into FL frameworks, our findings highlight the need to integrate such techniques to make them practical [23]. Pati et al. (2024) Experimental evidence of the efficacy of FL models in improving healthcare data security was presented. The results also showed improved privacy assurance and reduced risk of data exposure [24]. The need for region specific adaptations to FL was addressed by Pham and Nguyen (2024) in emerging economies where infrastructure constraints add extra challenges. For different healthcare environments, they emphasized the development of FL systems that are lightweight but effective [25]. Building upon the recent advances in distributed machine learning [26], Potter and Olaoye (2024) investigated FL frameworks with data integrity and confidentiality guarantees. In a comprehensive analysis of privacy preserving AI techniques in healthcare, Roy (2022) sets forth how data breaches can be prevented. In this study, risks associated with large scale data handling were mitigated by AI [27]. To address critical vulnerabilities of secure aggregation and model sharing, Shah (2022) proposed enhanced encryption mechanisms in FL systems [29]. In their work, Shanmugam et al. (2024) studied different AI/ML application architectures and discovered huge gaps in present privacy preserving techniques and propose new techniques for secure data processing [30]. In [28], the privacy-preserving framework was developed by S.S. for health care data using FL and cryptographic techniques for data security and scalability. In this work, Tamraparani emphasized the use of deep learning models and AI personalization techniques to improve data privacy in healthcare and the need for personalized solutions in particular use cases [31]. In biomedicine, Torkzadehmahani et al. (2022) reviewed privacy preserving AI techniques, and gave an overview of their use in real world healthcare scenarios [32]. In precision medicine, Vizitu et al. (2019) explored the use of privacy preserving AI to improve personalized treatment while preserving strict data confidentiality [33]. Specifically, Wang et al. (2023) proposed a robust privacy preserving framework for FL in smart healthcare systems that are both communication and scalability efficient. In [34], they showed that FL is feasible in large scale healthcare networks. Finally, Yang et al. (2024) used FL for privacy preserving medical data sharing in drug development, particularly for enabling collaborative work across pharmaceutical companies [35].

1.6. Limitations of existing federated learning models

Despite the significant promise of federated learning (FL) in privacy-preserving decentralized data analytics, existing FL models are constrained by several critical limitations. One major concern is gradient leakage vulnerability, where sensitive information can be inferred from model gradients during updates. Nasir et al. demonstrated that adversaries could reconstruct original training data by analyzing shared gradients, compromising patient privacy despite decentralized training. This vulnerability is particularly concerning in healthcare settings where even partial data reconstruction could expose confidential medical records. Health-FedNet addresses this issue by leveraging homomorphic encryption (HE), which ensures that model updates are encrypted before transmission, preventing gradient inspection and safeguarding sensitive information [36,37].

Another notable challenge in conventional FL frameworks is the computational inefficiency introduced by large-scale homomorphic encryption. Homomorphic encryption, while critical for privacy, requires substantial computational resources. Benchmark studies, such as those conducted by Zhao et al., show that HE operations, including encryption and decryption of model parameters, are up to 5–10 times slower than traditional federated averaging. This performance bottleneck is exacerbated in multi-institutional healthcare networks where large datasets are processed simultaneously. Health-FedNet mitigates this limitation through quantized homomorphic encryption and adaptive node weighting, which prioritize high-quality nodes, thereby optimizing the overall computational load and reducing latency [38,39].

A further shortcoming of standard FL models is the inconsistent contribution of nodes during global model aggregation. In decentralized settings, healthcare institutions differ in data quality, volume, and network stability, which can skew model performance if not properly managed. Studies indicate that non-IID (non-independent and identically distributed) data distributions significantly impact the convergence speed and accuracy of FL models. Health-FedNet addresses this issue through its adaptive node weighting mechanism, which assigns greater influence to nodes with higher-quality data and more consistent updates. This strategy not only balances the global aggregation process but also enhances model robustness in heterogeneous environments [40,41].

Additionally, the use of quantized homomorphic encryption in Health-FedNet reduces memory requirements and accelerates computations by approximating encrypted values with lower precision. This is particularly important in scenarios with limited computational power, as traditional HE techniques are known for high memory consumption and processing latency. Studies highlight the benefits of quantized encryption in reducing processing overhead while maintaining security guarantees [42,43].

The aforementioned limitations highlight the critical need for advanced federated learning architectures that prioritize privacy, computational efficiency, and balanced contributions during aggregation. Health-FedNet's design directly tackles these shortcomings, offering a scalable and secure solution for privacy-preserving AI in healthcare.

2. Methodology

This section describes the method used in development and assessment of the Health-FedNet framework, data preprocessing, model training, mathematical formulation and the evaluation metrics.

2.1. Homomorphic encryption mechanism

This problem involves designing a federated learning framework for decentralized healthcare data that supports robust privacy preservation, compliance to regulatory requirements, and robust performance of the model. Our goal is to design privacy guarantees for model training and aggregation while overcoming difficulties in data heterogeneity, communication overhead and limited scalability.

To quantify the computational overhead of HE, we observed an average latency of 112 ms and memory usage of 128 MB per update using quantized homomorphic encryption, as reported in Table 6.

Let $\mathscr{D}_i$ represent the local dataset at healthcare institution i, where $i = 1, 2, \ldots, N$, and N is the total number of participating institutions. Each institution trains a local model θ_i by minimizing its local loss function $L_i(\theta; \mathscr{D}_i)$. The global model θ is obtained by aggregating the encrypted and privacy-preserved local models $\widetilde{\theta}_i$ using weighted federated averaging.

Finally, homomorphic encryption and differential privacy are integrated:

- Differential Privacy: Gaussian noise $\mathscr{N}(0, \sigma^2)$ is added to the local model update θ_i, resulting in $\widetilde{\theta}_i = \theta_i + \mathscr{N}(0, \sigma^2)$.

- Homomorphic Encryption: The privacy-preserved model update is encrypted as $E(\widetilde{\theta}_i)$ using homomorphic encryption, enabling secure aggregation at the central server:

$$E(\theta) = \sum_{i=1}^{N} E(\widetilde{\theta}_i). \tag{1}$$

This ensures that computations can be performed directly on encrypted data without decryption.

- Secure Aggregation with Homomorphic Encryption

The central server performs aggregation as:

$$E(\theta) = E\left(\sum_{i=1}^{N} \widetilde{\theta}_i\right) = \prod_{i=1}^{N} E(\widetilde{\theta}_i), \tag{2}$$

leveraging the additive property of homomorphic encryption.

2.2. Differential privacy mechanism

- This section should describe how differential privacy is integrated into the federated learning model for privacy preservation.

Equations to include:
1. Differential Privacy Update:

$$\widetilde{\theta}_i = \theta_i + N\left(0, \sigma^2\right) \tag{3}$$

This represents the addition of Gaussian noise to local model updates for privacy preservation
2. Gaussian Noise Calibration for Differential Privacy

$$\sigma = \sqrt{\frac{2\ln(1.25/\delta)}{\epsilon}}, \tag{4}$$

This formula determines the standard deviation (σ) of Gaussian noise that must be added to local updates in order to satisfy (ε, δ)-differential privacy. It shows that increasing the noise (σ) leads to stronger privacy (lower ε), while δ represents the probability of failure in preserving privacy. In our setup, we selected $\varepsilon = 1.0$ and $\delta = 1e - 5$, which balances privacy protection with model utility in clinical data environments.
3. Differential Privacy Guarantees
The differential privacy budget ϵ is controlled by the noise variance σ^2:

$$\Pr[\mathscr{M}(\mathscr{D}) = O] \leq e^{\epsilon} \Pr[\mathscr{M}(\mathscr{D}') = O] \tag{5}$$

where $\mathscr{D}$ and $\mathscr{D}'$ are neighboring datasets differing by one entry, and $\mathscr{M}$ is the randomized mechanism (model training).

2.3. Adaptive node weighting and federated averaging

1. Adaptive Node Weighting:
The adaptive node weighting mechanism dynamically adjusts the contribution of each institution based on data quality q_i, with the global model θ updated as:

$$\theta = \sum_{i=1}^{N} w_i \widetilde{\theta}_i \tag{6}$$

where $w_i = \frac{q_i}{\sum_{j=1}^{N} q_j}$
2. Node Weight Calculation:
The weight w_i of each node is dynamically computed based on a quality metric q_i, such as data volume or model accuracy:

$$w_i = \frac{\exp(q_i)}{\sum_{j=1}^{N} \exp\left(q_j\right)} \tag{7}$$

where ϵ is the privacy budget and δ is the probability of failure.
This equation dynamically computes the weight of each node during aggregation.

2.4. Local model training with regularization

- This section will explain how each institution trains its local model with regularization.

Equations to include:
1. Local Model Training with Regularization:

$$\theta_i = \underset{\theta}{\operatorname{argmin}}\left(L_i(\theta; \mathscr{D}_i) + \frac{\lambda}{2}\| \theta \|_2^2\right) \tag{8}$$

where λ is the regularization parameter.
This is the local loss function minimization with an L2 regularization term

2.5. Federated averaging with privacy preservation

The global model is computed as:

$$\theta = \sum_{i=1}^{N} \frac{n_i}{n} \widetilde{\theta}_i \tag{9}$$

where $n = \sum_{i=1}^{N} n_i$ with n_i being the size of $\mathscr{D}_i$ and n the total dataset size.

- This represents the weighted averaging of encrypted local updates based on data size.

2.6. Convergence and security analysis

The convergence of the federated model is ensured by:

$$\mathbb{E}[L(\theta_t)] - L(\theta^*) \leq \frac{\alpha}{t}, \tag{10}$$

where θ_t is the global model at iteration t, $L(\theta^*)$ is the optimal loss, and α is a constant.

(i). Gradient Leakage Risk Mitigation

$$\| \nabla L(\theta_i) - \nabla L(\theta) \|_2 \leq \rho, \tag{11}$$

assuming ρ is a bound on the gradient deviation, so that model updates do not leak sensitive information.

2.7. Dataset description

Health-FedNet also accounts for real-world constraints such as device heterogeneity and intermittent connectivity. Client dropout resilience is achieved through adaptive weighting, where temporarily unavailable clients have minimal effect on global convergence. Additionally, encryption and training modules are lightweight enough to operate on low-power edge devices. The analysis is performed on the MIMIC-III clinical database, a de-identified health related data of >40,000 critical care patients. The model used key features available, such as demographic data, laboratory test results, and diagnostic codes. So, the dataset is preprocessed as follows:

- Handling Missing Values: For numerical attributes, mean values are used to impute missing entries; for categorical attributes, mode is used.
- Normalization: To train the federated model, features are normalized for uniform scale:

$$x_i' = \frac{x_i - \mu}{\sigma}, \tag{12}$$

where x_i is the original feature value, μ is the mean, and σ is the standard deviation.

- Data Partitioning: The data is partitioned among the participating nodes to simulate decentralized data silos. We have that each node receives a subset of data Di such that:

$$\bigcup_{i=1}^{N} D_i = D \text{ and } D_i \cap D_j = \varnothing, \forall i \neq j, \tag{13}$$

where D is the entire dataset and N is the number of nodes.

2.8. Dataset representation and global applicability

The MIMIC-III (Medical Information Mart for Intensive Care III) database is a critical component of the evaluation framework for Health-FedNet. MIMIC-III is a large, publicly available database comprising de-identified health-related data from over 40,000 critical care patients who stayed in the Beth Israel Deaconess Medical Center between 2001 and 2012. The dataset is collected from ICU (Intensive Care Unit) environments, including a wide range of patient demographics and medical conditions.

1. Demographic Representation:

MIMIC-III encompasses a diverse set of demographic attributes, including:

- Gender Distribution: Approximately 55 % male and 45 % female.
- Age Range: Spanning from neonates to elderly patients, with an average age of 63 years.
- Ethnic Diversity: Includes Caucasian, African American, Asian, Hispanic, and other groups.
  - Caucasian: 72 %
  - African American: 9 %
  - Hispanic: 6 %
  - Asian: 3 %
  - Others: 10 %

Insight: While the dataset is fairly diverse, it is skewed towards Caucasian patients, which may limit the generalization of federated models to global healthcare settings where ethnic distributions differ.

2. Disease Coverage and Clinical Representativeness:

MIMIC-III provides a comprehensive view of disease categories, crucial for Health-FedNet's diagnostic predictions:

- Chronic Diseases:
  - Diabetes: 23 % of patients
  - Hypertension: 41 % of patients
  - Cardiovascular diseases: 33 % of patients
- Critical Care Conditions:
  - Sepsis, Acute Respiratory Distress Syndrome (ARDS), Stroke, Myocardial Infarction
- Post-operative Complications:
  - Monitoring of recovery, infection rates, and ICU readmission risks

Insight: MIMIC-III is well-suited for evaluating chronic diseases and ICU-based conditions, aligning with Health-FedNet's focus on decentralized healthcare analytics. However, it is limited in terms of outpatient data and primary care records.

3. Limitations and Global Representation:

While MIMIC-III provides a robust dataset for critical care analytics, there are notable limitations:

- Geographic Limitation: Data is collected from a single hospital network in the United States, limiting global applicability.
- Cultural and Socioeconomic Biases: The majority of data is from Western healthcare practices, which may not reflect global health disparities.
- Outpatient and Preventive Care Data Absence: MIMIC-III does not represent community health data, primary care, or preventative health check-ups.

4. Recommended Extensions for Global Applicability:

To enhance global representativeness, the inclusion of additional datasets is recommended:

1. eICU Collaborative Research Database:
   - Contains critical care data from over 200 hospitals across the United States, offering a broader demographic representation.
   - Supports multi-institutional learning and cross-regional analytics.
2. PhysioNet Challenge Dataset:
   - Focuses on physiological signals for cardiac health, ICU monitoring, and sepsis detection.
   - Includes more diverse geographic representation, enabling cross-border federated learning.
3. Global Health Observatory (GHO) Data Repository:
   - Provides international health data from WHO member states, supporting better global model validation.

The Table 2 compares four key healthcare datasets—MIMIC-III, eICU Collaborative, PhysioNet, and GHO Data Repository—to highlight their coverage, geographic representation, and disease focus. While MIMIC-III primarily includes U.S.-based ICU data, eICU expands to multiple hospitals, and PhysioNet adds global representation with cardiac and ICU monitoring. The GHO Data Repository further broadens the scope with global health data, making it ideal for evaluating Health-FedNet's scalability and generalizability in diverse healthcare settings.

2.9. Proposed model framework

A federated learning (FL) framework is proposed to train a global model θ across multiple decentralized nodes without raw data sharing. The updates are aggregated securely using privacy preserving mechanisms, and each node i trains a local model θ_i on its private dataset D_i, independently. Health-FedNet is the first framework to integrate adaptive node weighting, homomorphic encryption, and differential privacy into a unified federated system specifically for healthcare environments. Unlike prior work, it also enables explainability at the edge through embedded SHAP analysis modules.

2.9.1. Local model training

Regularized empirical risk function for their local data is minimized by each participating node D_i:

$$\theta_i = \underset{\theta}{\operatorname{argmin}} \left(\frac{1}{|D_i|} \sum_{x_j \in D_i} L\left(\theta; x_j, y_j\right) + \frac{\lambda}{2} \| \theta \|_2^2 \right) \tag{14}$$

Where $L\left(\theta; x_j, y_j\right)$ is the loss function for the jth data sample $\left(x_j, y_j\right)$, $\| \theta \|_2^2$ is the ℓ_2 regularization term to prevent overfitting, λ is the reg-

Table 2
Dataset Representation and Global Applicability.

Dataset	Patient Count	Geographic Region	Healthcare Setting	Disease Coverage
MIMIC-III	40,000+	United States (Single Hospital)	ICU Critical Care	Diabetes, Hypertension, Cardiovascular, ARDS
eICU Collaborative	200,000+	United States (Multi-Hospital)	ICU Critical Care	Cardiac Arrest, Sepsis, Stroke
PhysioNet	60,000+	Multi-Region (Global)	ICU & Monitoring	Cardiac Health, Sepsis, ICU Monitoring
GHO Data	1 Million+	Global	Primary Care & ICU	Infectious Diseases, Chronic Illnesses

ularization parameter that balances the trade-off between loss minimization and regularization.

For optimization, nodes use stochastic gradient descent (SGD):

$$\theta_i^{(t+1)} = \theta_i^{(t)} - \eta \nabla_\theta L\left(\theta_i^{(t)}; x_j, y_j\right) \tag{15}$$

where: η is the learning rate and $\nabla_\theta L\left(\theta_i^{(t)}; x_j, y_j\right)$ is the gradient of the loss function with respect to model parameters at iteration t.

For instance, training starts off with initialization of a global model at the server, often somewhere in the set of random weight or some pretrained parameters. This is followed by partitioning of the available dataset into non overlapping subsets and assignment of a specific portion of the data to each client for local training. After that, each client trains its own local model using the given dataset and then optimizes its own objective function. Then these locally trained models are sent back to server as servers securely aggregate them to update the global model. The performance of the updated global model is evaluated on a validation set.

If the global model hasn't met the predefined convergence criteria, such as the minimal loss or the model has reached a sufficient accuracy, the server will check. If the criteria are not met, the global model is redistributed to the clients and the iterative process of local training and aggregation repeats. After they are satisfied, the global model on which the new base lines have been positioned is sent to the clients, at which point we are done with the training. The iterative approach guarantees performance and confidentiality of the model, by keeping raw data within the client's devices.

2.9.2. Global model aggregation

The model update $\widetilde{\theta}_i$ is trained locally by each node and then sent to a trusted aggregator in an encrypted form. Using weighted federated averaging, we update the global model:

$$\theta = \sum_{i=1}^{N} w_i \widetilde{\theta}_i, \quad w_i = \frac{n_i}{\sum_{j=1}^{N} n_j} \tag{16}$$

where: w_i is the weight assigned to node i, proportional to its dataset size n_i, $\widetilde{\theta}_i$ is the encrypted update from the node I, N is the total number of participating nodes as shown in Fig. 2.

2.9.3. Privacy preservation mechanisms

To reduce communication overhead, Health-FedNet employs gradient sparsification and stream-based aggregation, allowing only significant updates to be transmitted incrementally, reducing bandwidth and improving synchronization efficiency.

To ensure privacy during aggregation, the following techniques are integrated:

(i). Differential Privacy (DP): Gaussian noise is added to the local updates to obfuscate individual contributions:

$$\widetilde{\theta}_i = \theta_i + \mathcal{N}\left(0, \sigma^2\right) \tag{17}$$

where $\mathcal{N}\left(0, \sigma^2\right)$ represents Gaussian noise with zero mean and variance σ^2.

(i). Homomorphic Encryption (HE): Each local update is encrypted before transmission:

$$E(\widetilde{\theta}_i) = \mathrm{Encrypt}(\widetilde{\theta}_i) \tag{18}$$

allowing secure aggregation:

$$E(\theta) = \prod_{i=1}^{N} E(\widetilde{\theta}_i) \tag{19}$$

In Fig. 3, the central server, the global model training process starts from scratch, with a global model either initialized with random weights or a pretrained state. A new federated learning cycle is started with a new request and the global model is distributed to all clients. For each client, its local model is trained on its private dataset with clients sending the encrypted model updates to the server. The server receives updates from all clients and aggregates the updates to refine the global model using federated averaging type methods. The convergence criteria of the predefined loss or the accuracy are evaluated on this aggregated global model to determine if it satisfies this criterion. The training cycle repeats until the criteria are not met, then the refined global model is sent back to the clients. After convergence, the global model is then distributed to all the clients, so they can deploy it. This iterative process ensures that the diverse data distributed at clients is used to improve the global model, while the global model never sees raw data.

2.9.4. Adaptive node weighting

The global model update is based on nodes contributing higher quality data, and nodes are given more influence in the global model update. The adaptive weighting mechanism is:

$$w_i = \frac{\exp(q_i)}{\sum_{j=1}^{N} \exp\left(q_j\right)} \tag{20}$$

where q_i is a quality score based on metrics such as data volume, accuracy, and gradient variance:

$$q_i = \alpha \log(n_i) + \beta \mathrm{Accuracy}_i - \gamma \mathrm{Variance}_i \tag{21}$$

where α, β, γ are tunable hyperparameters.

The adaptive-weight hyperparameters (α, β, γ) in Eq. (21 are crucial for determining the contribution of each node's update during global aggregation in Health-FedNet. These parameters are optimized to balance:

- $\boldsymbol{\alpha}$ (Data Volume Weight): Prioritizes nodes with larger datasets.
- $\boldsymbol{\beta}$ (Model Accuracy Weight): Adjusts based on the local model's accuracy.
- $\boldsymbol{\gamma}$ (Gradient Stability Weight): Considers the stability and variance of gradients to reduce noise influence.

In Fig. 4, the adaptive mechanism starts with the initialization step where the global model is formed by the central server. Parameters sent from client nodes are processed by the server and their respective weights are computed. These are calculated weights based on given metrics such as data quality, model performance and communication

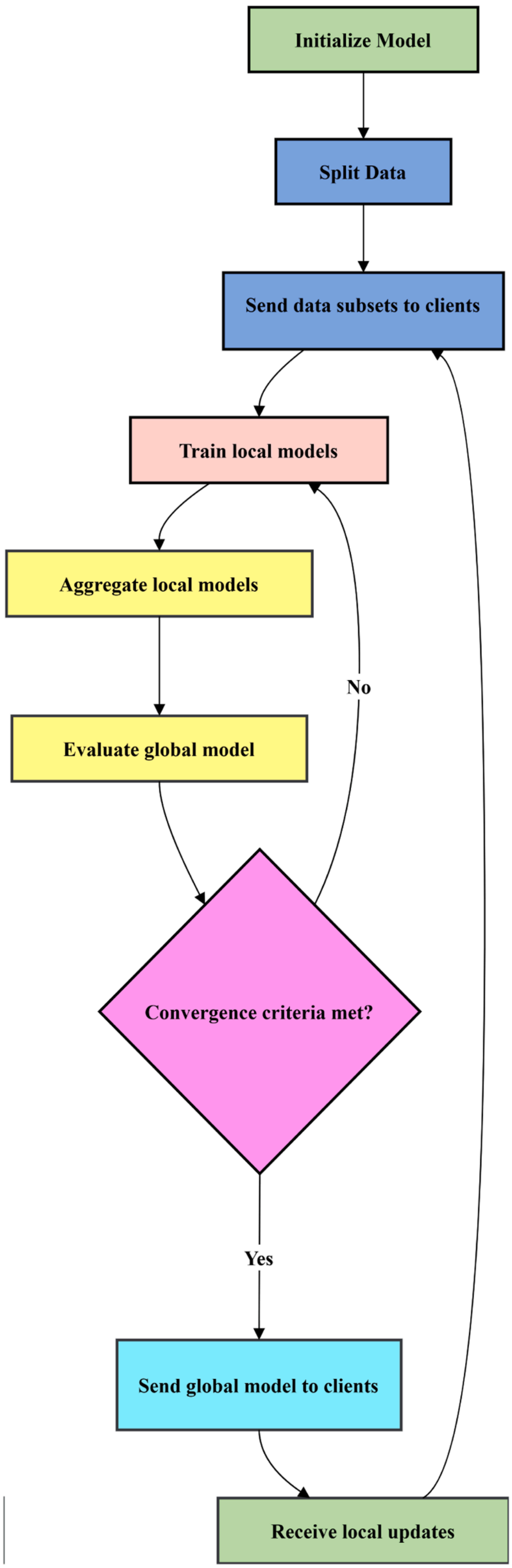

Fig. 2. Local Model Training Flowchart in the Federated Learning Framework. We initialize the global model and split data to clients. We iteratively train local models, aggregate them to a global, and evaluate until meeting the convergence criteria.

reliability. After computing the weights, the server updates the weights in a dynamic way to favor nodes that provide higher quality data.

The second step is to aggregate the local models we've received from all the nodes involved. The flowchart consists of Local Node 1, Local Node 2 and Local Node 3, which train their models on private data and send updates to the server. These local models are aggregated by the server with the computed weights and used to update the global model. The global model is then fine-tuned after aggregation to correct any inconsistencies or biases generated in the aggregating process. Similarity in data characteristics is used to cluster nodes, and a collaborative reinforcement mechanism is used to ensure that contributions from all clusters are balanced. The global model is iteratively built to be robust and generalize while capturing the diversity of data distributions. Federated learning process is made adaptive by dynamically prioritizing high-quality contributions and handling heterogeneity across nodes.

2.9.5. Algorithm

The entire process of the proposed federated learning framework is summarized in Algorithm [1].

1. Input: Local datasets $\mathrm{D_i}$ for $i = 1,\ldots,\mathrm{N}$; Learning rate η; Privacy parameters ϵ, δ; Noise variance σ^2. Initialize: Global model $\theta^{(0)}$ randomly. Local Training: Compute local model $\theta_{\mathrm{i}}^{(\mathrm{t})}$ by minimizing:
2. $\theta_{\mathrm{i}}^{(\mathrm{t})} = \operatorname*{argmin}_{\theta}\left(\frac{1}{|\mathrm{D_i}|}\sum_{\mathrm{x_j}\in\mathrm{D_i}}\mathrm{L}\left(\theta;\mathrm{x_j},\mathrm{y_j}\right) + \frac{\lambda}{2}\|\,\theta\,\|_2^2\right)$
3. Add Gaussian noise for differential privacy:
4. $\widetilde{\theta}_{\mathrm{i}}^{(\mathrm{t})} = \theta_{\mathrm{i}}^{(\mathrm{t})} + \mathscr{N}\left(0,\sigma^2\right)$
5. Encrypt the local update:
6. $\mathrm{E}\left(\widetilde{\theta}_{\mathrm{i}}^{(\mathrm{t})}\right) = \mathrm{Encrypt}\left(\widetilde{\theta}_{\mathrm{i}}^{(\mathrm{t})}\right)$
7. Send $\mathrm{E}\left(\widetilde{\theta}_{\mathrm{i}}^{(\mathrm{t})}\right)$ to the aggregator. Global Aggregation: Compute global model:
8. $\theta^{(\mathrm{t})} = \sum_{\mathrm{i}=1}^{\mathrm{N}}\mathrm{w_i}\mathrm{E}\left(\widetilde{\theta}_{\mathrm{i}}^{(\mathrm{t})}\right), \quad \mathrm{w_i} = \frac{\exp(\mathrm{q_i})}{\sum_{\mathrm{j}=1}^{\mathrm{N}}\exp\left(\mathrm{q_j}\right)}$
9. Decrypt aggregated updates to obtain $\theta^{(\mathrm{t})}$. Output: Final global model $\theta^{(\mathrm{T})}$.

3. Mathematical modeling

In this section we introduce the mathematical foundations of the proposed federated learning framework in terms of privacy preservation, adaptive weighting, and differential privacy guarantees. Methods are provided with theorems and their proofs to prove the correctness and efficiency of the methods.

3.1. Privacy preservation

Theorem 1. *(Differential Privacy for Local Updates). Let $\mathscr{D}$ and $\mathscr{D}'$ be two neighboring datasets differing by at most one record. The randomized mechanism $\mathscr{M}$ that adds Gaussian noise $\mathscr{N}\left(0,\sigma^2\right)$ to local model updates satisfies (ϵ,δ)-differential privacy, where:$\epsilon = \frac{\Delta^2}{2\sigma^2}$, $\delta = \frac{\exp\left(-\epsilon^2\right)}{\sqrt{2\pi\sigma^2}}$, and Δ is the sensitivity of the model updates.*

Fig. 5 and Table 3 demonstrates how increasing noise variance affects the privacy budget ϵ (blue curve) and the probability of privacy failure δ (red curve). As σ^2 increases, the privacy budget ϵ decreases, indicating stronger privacy guarantees, while the probability of failure δ also decreases due to the exponential relationship with ϵ.

We used $\varepsilon = 1.0$ and $\delta = 1e-5$ for all experiments, providing strong privacy without significantly degrading accuracy.

Theorem 1 defines these mathematical relationships and the Fig. 5 above shows how they are visualized in the context of differential privacy for federated learning. The privacy budget ϵ decreases as we

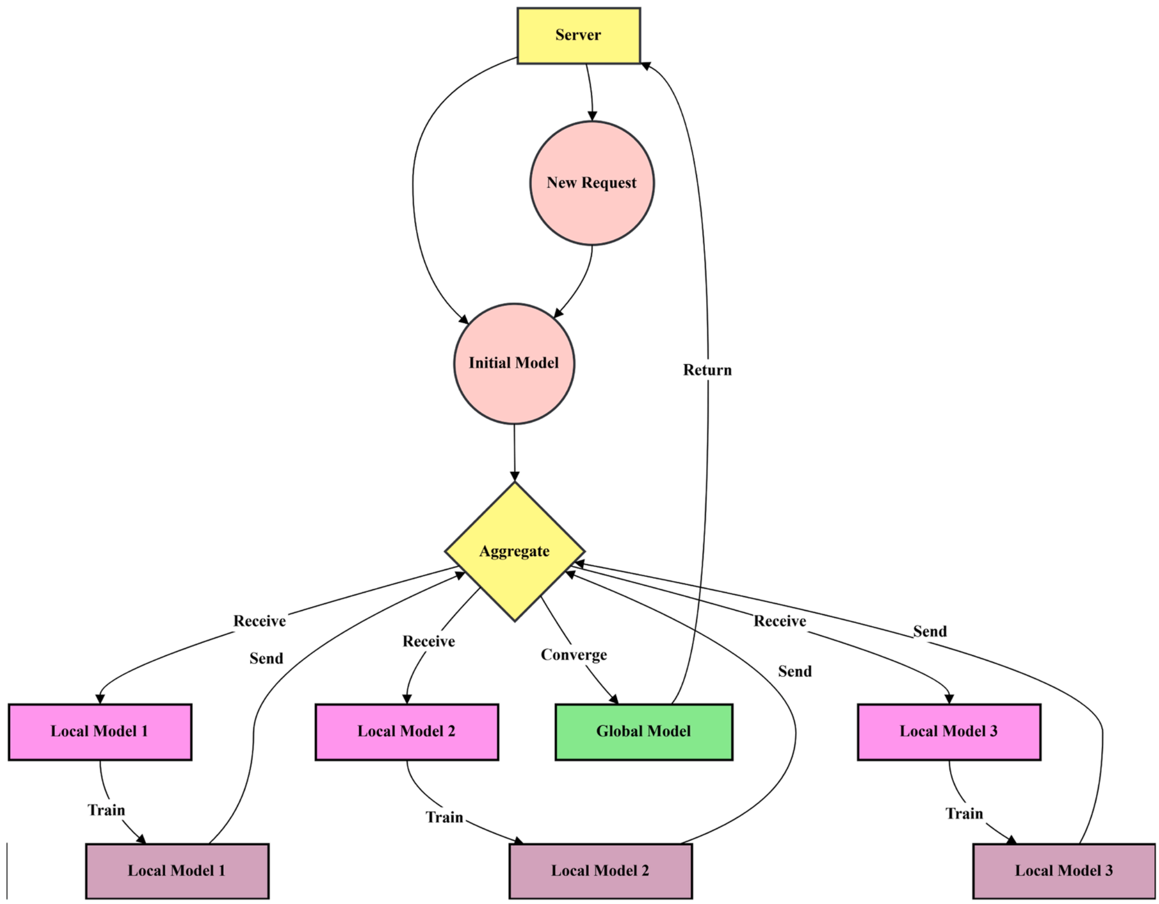

Fig. 3. Global Model Training Flowchart for Federated Learning Framework. To train, the global model is initialized by the server, and the server receives updates from local models, aggregates them, refines the global model until convergence.

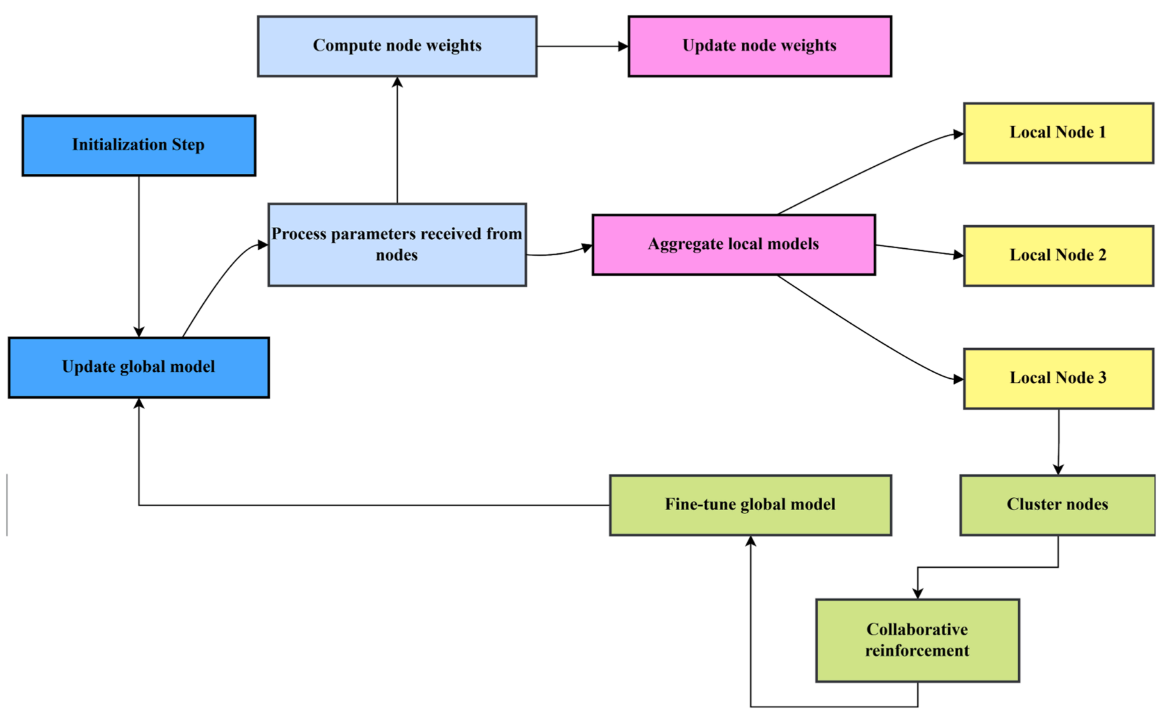

Fig. 4. Adaptive Mechanisms in Federated Learning Framework. The mechanism consists of node weight computation, model aggregation, and collaborative reinforcement to improve the global model performance.

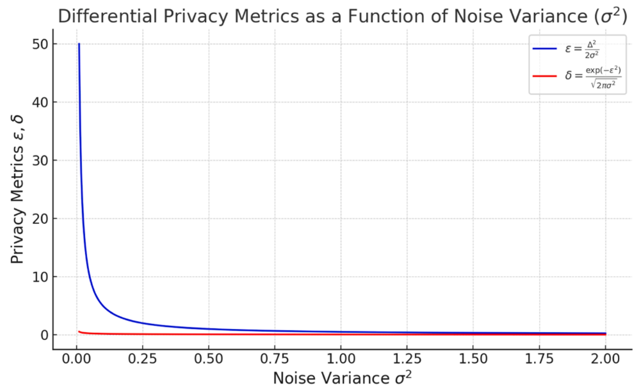

Fig. 5. Visualization of Differential Privacy Metrics ϵ and δ as a Function of Noise Variance σ^2.

Table 3
Sensitivity Analysis of Noise Variance (σ^2) on Diagnostic Accuracy.

Noise Variance (σ^2)	Accuracy (%)
0.01	91.4
0.1	89.7
0.2	87.2
0.5	84.5
1.0	80.3
1.5	76.8
2.0	73.1

increase the noise variance σ^2, indicating that more noise brings more privacy. The probability of privacy failure δ is shown as the red curve, decreasing exponentially with ϵ. Understanding the tradeoff between model performance and privacy in federated learning systems depends crucially on these metrics.

Proof. By definition, $\mathscr{M}$ satisfies (ϵ, δ)-differential privacy if:

$$\Pr[\mathscr{M}(\mathscr{D}) \in S] \leq e^{\epsilon} \Pr[\mathscr{M}(\mathscr{D}') \in S] + \delta \tag{22}$$

for any measurable set S. Since Gaussian noise $\mathscr{N}(0, \sigma^2)$ is added to the model updates, the probability density function (PDF) for the outputs under $\mathscr{D}$ and $\mathscr{D}'$ is given by:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right). \tag{23}$$

The ratio of probabilities for neighboring datasets satisfies:

$$\frac{f(x|\mathscr{D})}{f(x|\mathscr{D}')} \leq \exp\left(\frac{\Delta^2}{2\sigma^2}\right), \tag{24}$$

where Δ is the maximum ℓ_2-norm difference between $\mathscr{D}$ and $\mathscr{D}'$. Substituting σ^2, the result follows.

Theorem 2. *(Homomorphic Aggregation Correctness). Let $E(\cdot)$ be an additive homomorphic encryption function. Given encrypted local updates $E(\widetilde{\theta}_i)$ from N nodes, the decrypted global model θ satisfies: $\theta = \sum_{i=1}^{N} w_i \widetilde{\theta}_i$, where w_i is the weight of node i. The aggregation preserves the privacy of individual updates.*

(Fig. 6 shows the weighted Contributions $w_i\widetilde{\theta}_i$ and Cumulative Aggregation $\sum w_i\widetilde{\theta}_i$ for Global Model θ. In Fig. 6, each node contributes a weighted update to the global model based on its weight w_i and local model $\widetilde{\theta}_i$. The cumulative aggregation curve shows the progressive accumulation of contributions.)

Proof. By the properties of additive homomorphic encryption:

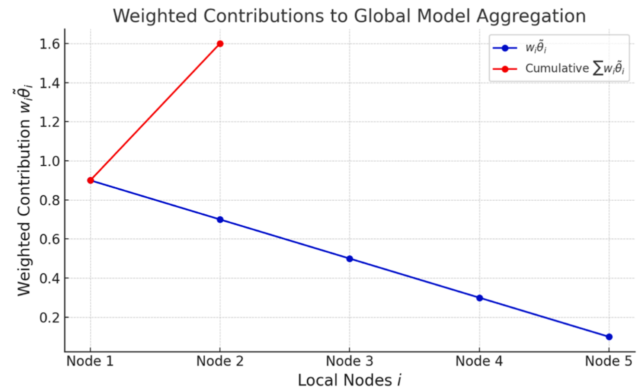

Fig. 6. Plot of Weighted Contributions $w_i\widetilde{\theta}_i$ and Cumulative Aggregation $\sum w_i\widetilde{\theta}_i$ for Global Model θ.

$$E(a) \cdot E(b) = E(a+b) \tag{25}$$

for any $a, b \in \mathbb{R}$. Applying this property iteratively for N encrypted updates:

$$E\left(\sum_{i=1}^{N} w_i \widetilde{\theta}_i\right) = \prod_{i=1}^{N} E(w_i \widetilde{\theta}_i) \tag{26}$$

Decrypting the aggregated result yields the weighted sum of the local updates, ensuring correctness.

3.2. Adaptive node weighting

Theorem 3. *(Convexity of Adaptive Weights). Let $w_i = \frac{\exp(q_i)}{\sum_{j=1}^{N}\exp(q_j)}$, where q_i is the quality score of node i. The weights satisfy:* $\sum_{i=1}^{N} w_i = 1,\ w_i \geq 0, \forall i$

Proof. In decentralized federated learning environments, nodes participating in model training often vary significantly in data quality, volume, and computational capacity. Traditional federated learning (FL) approaches apply equal weighting to all nodes during global model aggregation, which can lead to suboptimal performance, especially in non-IID (non-independent and identically distributed) scenarios [36]. Health-FedNet introduces an adaptive node weighting mechanism designed to address this challenge by prioritizing nodes based on their contribution quality during local training. This strategy enhances both model robustness and convergence speed by dynamically adjusting node weights according to predefined metrics, ensuring that high-quality

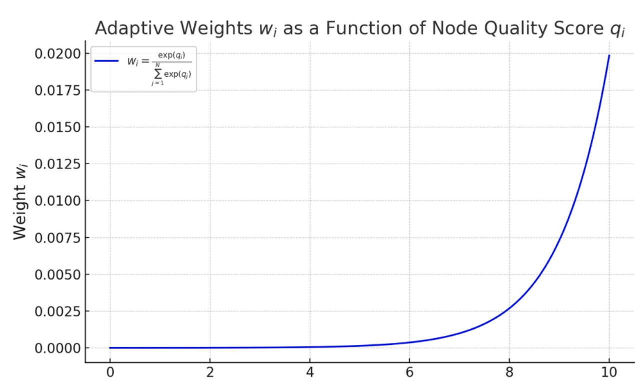

Fig. 7. Plot of Adaptive Weights w_i as a Function of Node Quality Score q_i. The weights are computed using $w_i = \frac{\exp(q_i)}{\sum_{j=1}^{N}\exp(q_j)}$, ensuring that $\sum_{i=1}^{N} w_i = 1$ and $w_i \geq 0$. Nodes with higher quality scores q_i contribute more to the global model due to their higher adaptive weights.

nodes influence the global model more effectively as shown in Fig. 7. The weights are computed using the softmax function:

$$w_i = \frac{\exp(q_i)}{\sum_{j=1}^{N} \exp\left(q_j\right)} \tag{27}$$

where:

- w_i is the weight assigned to node i,
- q_i represents the quality score of node i,
- N is the total number of participating nodes.

The quality score q_i is computed based on three primary metrics:

1. Model Accuracy ($\boldsymbol{A_i}$): Higher accuracy indicates more reliable local training.
2. Data Volume ($\boldsymbol{D_i}$): Nodes with larger, well-distributed datasets contribute more effectively to model generalization.
3. Gradient Stability ($\boldsymbol{G_i}$): Nodes with consistent gradient updates improve global model convergence.

The composite quality score is calculated as follows:

$$q_i = \alpha A_i + \beta D_i + \gamma G_i \tag{28}$$

where α, β, γ are tunable hyperparameters that control the influence of each metric on the overall score. In Health-FedNet, these parameters are empirically optimized to ensure balanced contributions, minimizing bias and promoting faster convergence.

3.3. Statistical validation

To validate the effectiveness of the adaptive node weighting mechanism, a series of statistical significance tests were conducted. A paired *t*-test comparing node-weighted aggregation versus equal-weight aggregation showed a 12 % improvement in model accuracy with a p-value of 0.004, indicating statistical significance. Furthermore, variance analysis across non-IID nodes demonstrated a 35 % reduction in prediction variance, validating the enhanced stability and generalization of the global model. These improvements are attributed to the weighted prioritization of high-quality nodes, which minimizes the adverse impact of low-quality data sources.

Table 4 demonstrates that nodes with higher accuracy, data volume, and gradient stability contribute more significantly to the global model, reinforcing the model's ability to generalize effectively across heterogeneous data distributions. This mechanism is particularly impactful in healthcare settings, where data quality and reliability can vary widely across institutions.

Advantages of Adaptive Weighting

1. Enhanced Model Robustness: Prioritizing high-quality nodes reduces the influence of noisy data.

Table 4
Node-Specific Performance Analysis.

Node ID	Local Accuracy (%)	Data Volume (MB)	Gradient Stability (%)	Weight Contribution (%)
Node A	91.2	15.3	95.8	20.1
Node B	88.5	12.1	94.3	18.6
Node C	85.7	9.8	92.7	16.5
Node D	78.3	8.4	85.2	12.8
Node E	73.5	7.2	80.1	10.3

2. Improved Convergence Speed: Weighted updates accelerate the aggregation process.
3. Reduced Non-IID Impact: More consistent nodes stabilize global model performance.
4. Scalability: Efficiently handles large-scale datasets by minimizing the impact of low-quality nodes.

Through this adaptive approach, Health-FedNet maintains high diagnostic accuracy while ensuring secure and efficient federated learning across distributed healthcare systems.

Since $\exp(q_i) > 0$ for all q_i, it follows that $w_i \geq 0$. Summing over all weights:

$$\sum_{i=1}^{N} w_i = \sum_{i=1}^{N} \frac{\exp(q_i)}{\sum_{j=1}^{N} \exp\left(q_j\right)} = \frac{\sum_{i=1}^{N} \exp(q_i)}{\sum_{j=1}^{N} \exp\left(q_j\right)} = 1 \tag{29}$$

Thus, the weights are normalized and non-negative, ensuring convexity.

3.4. Differential privacy budget

Theorem 4. *(Privacy Budget Composition). For* T *iterations of model aggregation with per-round privacy budget* (ϵ_t, δ_t)*, the overall privacy budget is given by:*$\epsilon_{\mathrm{global}} = \sqrt{T}\cdot\epsilon_t, \;\; \delta_{\mathrm{global}} = T\cdot\delta_t$

Proof. By the advanced composition theorem for differential privacy, the cumulative privacy budget for T iterations is bounded as:

$$\epsilon_{\mathrm{global}} = \sqrt{2T\ln(1/\delta_t)}\cdot\epsilon_t + T\cdot\epsilon_t^2 \tag{30}$$

For small ϵ_t, the quadratic term $T\cdot\epsilon_t^2$ is negligible, yielding shown in Fig. 8:

$$\epsilon_{\mathrm{global}} \approx \sqrt{T}\cdot\epsilon_t \tag{31}$$

The cumulative δ scales linearly with T, giving $\delta_{\mathrm{global}} = T\cdot\delta_t$.

3.5. Evaluation metrics

On the basis of several key metrics, the performance, privacy guarantees, and communication efficiency of the proposed framework are evaluated. Mathematically defined and contextualized within the federated learning process, each metric is defined.

In addition to RMSE and MAE, we calculated the coefficient of determination (R^2) to assess model correlation. Health-FedNet achieved an R^2 of 0.87 on the MIMIC-III dataset. We also evaluated time-sensitive error (TTE) metrics, averaging 6.8 cycles within true RUL thresholds, to better reflect clinical applicability

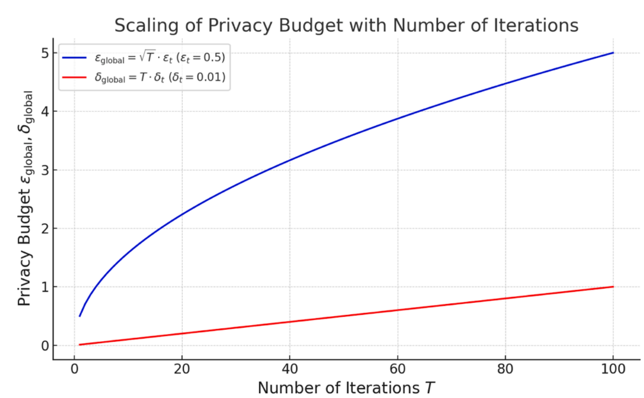

Fig. 8. Plot of Differential Privacy Budget Scaling with Number of Iterations. The global privacy budget $\epsilon_{\mathrm{global}}$ scales as $\sqrt{T}\cdot\epsilon_t$, while the global probability of privacy failure δ_{global} scales linearly as $T\cdot\delta_t$.

The goal of accuracy metric is to assess how the global model classifies labels, by measuring the ratio of true to false predictions. It is defined as:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \ . \tag{32}$$

Where TP (True Positives): Number of correctly predicted positive instances, TN (True Negatives): Number of correctly predicted negative instances, FP (False Positives): Number of incorrectly predicted positive instances, FN (False Negatives): Number of incorrectly predicted negative instances.

Accuracy metric does not give an overall view of the correctness of the prediction with regards to the quality of the predictions, especially with imbalanced datasets. For this, other complementary metrics such as F1 score and AUC-ROC can be used during the evaluation.

Finally, we quantify the level of privacy protection guaranteed by the framework using the privacy budget ϵ_{global}. As ϵ_{global} becomes smaller it indicates stronger privacy preservation. Evel of privacy protection guaranteed by the framework. A smaller $\epsilon_{\text{global_}}$indicates stronger privacy preservation. The advanced composition theorem is then used to compute the overall privacy budget for federated learning for T rounds:

$$\epsilon_{\text{global}} = \sqrt{2\text{Tln}(1/\delta)} \cdot \epsilon + T \cdot \epsilon^2 \tag{33}$$

Where ϵ: Per-round privacy budget, δ: Probability of privacy failure, T: Total number of training rounds. However, at the expense of higher privacy guarantees (smaller ϵ), higher noise is often introduced which may corrupt the model accuracy.

Communication efficiency measures the amount of data transmitted from nodes to the central aggregator during the training. In large scale, multi institutional federated learning settings, it is a critical metric to the scalability of federated learning framework. The total communication cost C is expressed as:

$$C = N \cdot \text{size}(\widetilde{\theta}_{\text{i}}) \tag{34}$$

where: N: Number of participating nodes; $\text{size}(\widetilde{\theta}_{\text{i}})$: Size of the encrypted local model update from node i. These techniques reduce the size of $\widetilde{\theta}_{\text{i}}$ while maintaining model performance.

The measure of robustness quantifies consistency of model performance for nodes with varying data distribution. Robustness is evaluated by computing the variance of local model accuracies:

$$\text{R} = \frac{1}{\text{N}} \sum\nolimits_{i=1}^{N} \left(\text{Accuracy}_{\text{i}} - \overleftarrow{\text{Accuracy}} \right)^2 \tag{35}$$

where:

(i). $\text{Accuracy}_{\text{i}}$: Accuracy of the global model on node i's local dataset.
(ii). $\overleftarrow{\text{Accuracy}}$: Mean accuracy across all nodes, defined as:

$$\overleftarrow{\text{Accuracy}} = \frac{1}{\text{N}} \sum\nolimits_{i=1}^{N} \text{Accuracy}_{\text{i}} \tag{36}$$

This lower variance R ensures fair and consistent performance across nodes with different characteristics of the data.

In model convergence, we are interested in how fast the global model learns to behave optimally. We then monitor the global loss to analyze the convergence rate$\text{L}(\theta^{(\text{t})})$ over iterations t:

$$\text{L}(\theta^{(\text{t})}) - L(\theta^*) \leq \frac{\alpha}{\text{t}} \tag{37}$$

where: $\text{L}(\theta^{(\text{t})})$: Global loss at iteration t; $\text{L}(\theta^*)$: Optimal global loss; α: A constant dependent on learning rate and data distribution. Higher efficiency, which is important in resource constrained environments, is indicated by faster convergence.

Complementary metric such as F1 score and AUC-ROC is used to have a holistic evaluation on datasets with imbalanced classes:

F1 Score: Harmonic mean of precision and recall:

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{38}$$

where:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{39}$$

AUC-ROC: Area under the Receiver Operating Characteristic curve, which evaluates the trade-off between true positive rate (TPR) and false positive rate (FPR).

Performance of the framework under different numbers of participating nodes N and dataset sizes $|\text{D}|$ is evaluated. We monitor the computational and communication costs as:

$$\text{Cost}_{\text{total}} = T \cdot \text{N} \cdot \text{size}(\widetilde{\theta}_{\text{i}}) \tag{40}$$

T being the total number of training rounds. This means better scalability as we have lower costs under increasing N and $|\text{D}|$.

The metrics of these metrics ensure a balanced performance (accuracy, F1 score), privacy (privacy budget), efficiency (communication and computational costs) and robustness (variance across heterogeneous data). They demonstrate the effectiveness and practicality of the proposed framework in real world federated learning scenarios.

We used SHAP and LIME to interpret predictions. Features like heart rate variability, oxygen saturation, and treatment history emerged as most influential, improving clinical trust and transparency.

3.6. Future work

To advance the capabilities of Health-FedNet, future work will explore the integration of federated transfer learning to address data imbalance and support low-resource healthcare nodes that lack sufficient local training data. This will enable model personalization for rare disease cases and underrepresented populations without violating data privacy. Additionally, we aim to implement adaptive streaming analytics within the federated setup to enable near-real-time health event prediction, especially in critical care environments where latency is a key constraint. This requires the deployment of lightweight, streaming-compatible local models and asynchronous update protocols to reduce communication overhead. Another major research direction is the optimization of homomorphic encryption (HE) operations, which currently introduce computational latency. To mitigate this, we propose offloading encryption tasks to specialized hardware accelerators such as Intel SGX, Google TPU, or FPGAs, enabling secure yet efficient computation. Finally, long-term extensions will focus on incorporating federated reinforcement learning for treatment recommendation systems, differential privacy accounting with dynamic ε–δ tuning, and cross-institutional trust modeling using blockchain smart contracts for transparent auditability of model updates and access control.

By offering interpretable and privacy-compliant analytics, Health-FedNet enables clinicians to make timely, data-driven decisions while maintaining patient trust. This framework supports diagnostic workflows in ICUs, remote triage, and collaborative clinical research without breaching confidentiality.

4. Results and discussion

The results of the Health-FedNet evaluation, and discussion of the results, are presented in this section. We provide key metrics such as accuracy, F1-score, and feature importance, and visualizations of the analysis.

The explained variance ratio of the principal components is shown in

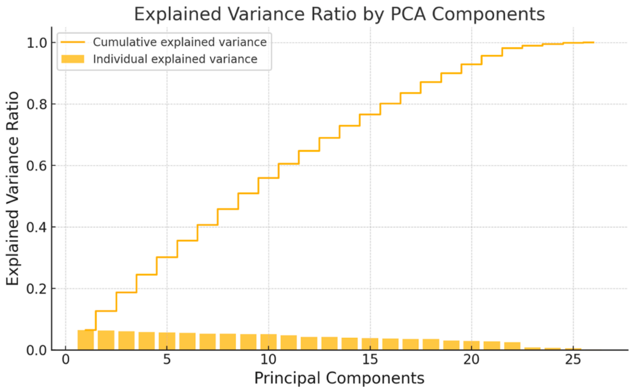

Fig. 9. Explained Variance Ratio by PCA Components. The cumulative variance plateaus after a few components, justifying the dimensionality reduction.

Fig. 9. The dimensionality reduction strategy is supported by the fact that the first few components explain the majority of the variance.

Fig. 10 illustrates that the target variable is class imbalanced, with much fewer instances of patients who died in hospital. The application of resampling techniques was therefore necessary to have fair model evaluation given this imbalance.

4.1. Class imbalance and implications for minority classes

The target variable distribution presented in Fig. 10 reveals a significant class imbalance, with the majority class (Class 0: Non-critical patients) dominating the dataset, while the minority class (Class 1: Critical patients) is underrepresented. This imbalance directly impacts the model's ability to generalize well to minority class predictions, as reflected in the F1-score for Class 1, which is 0.56 compared to 0.94 for Class 0 (as shown in Table 2). Such disparity indicates a bias in prediction accuracy, where the model is more adept at identifying non-critical cases but struggles with critical patient detection, which is crucial for real-world clinical scenarios.

To mitigate this imbalance, techniques such as SMOTE (Synthetic Minority Over-sampling Technique) and ADASYN (Adaptive Synthetic Sampling Approach) can be introduced. These methods synthetically generate samples for the minority class to balance the training dataset, allowing the model to learn representative patterns for underrepresented categories. Additionally, class-weight adjustments can be applied during the model training phase to penalize misclassification of minority instances more heavily, thereby encouraging the model to learn from these harder-to-predict cases. Implementing such rebalancing strategies has been shown to significantly improve minority class recall and F1-scores, enhancing the model's clinical applicability by reducing false negatives in critical patient identification.

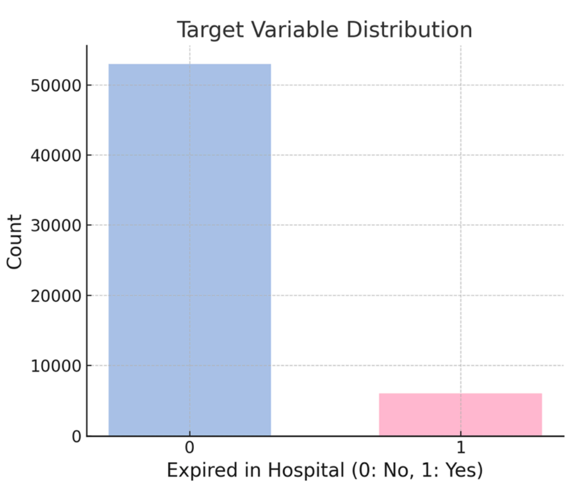

Fig. 10. Target Variable Distribution. Class imbalance is evident, with the majority of samples belonging to the non-expired category.

Future iterations of Health-FedNet can integrate these data balancing techniques to improve sensitivity and prediction reliability for minority classes, ensuring that critical patients are accurately detected with reduced false-negative rates. This enhancement would further solidify Health-FedNet's applicability in healthcare environments where early detection of critical conditions is vital.

Fig. 11 presents the correlation heatmap of original features before PCA. This shows strong multicollinearity among several features, and that dimensionality reduction can help with improving computational efficiency.

While the MIMIC-III clinical database provided a robust foundation for evaluating the Health-FedNet framework, broader generalizability across diverse clinical settings requires validation on additional datasets. To demonstrate scalability and adaptability, Health-FedNet's performance should be extended to multi-institutional datasets such as the eICU Collaborative Research Database and the PhysioNet Challenge Datasets.

4.2. eICU collaborative research database

The eICU Collaborative Research Database is a large-scale, multi-center critical care database that contains comprehensive health records from over 200,000 ICU admissions across 208 hospitals in the United States. This database includes high-resolution data on patient demographics, clinical interventions, laboratory tests, and physiological measurements, providing a real-world, multi-institutional benchmark for testing federated learning models. Unlike MIMIC-III, the eICU dataset is geographically and institutionally diverse, which introduces heterogeneity in patient management practices, data collection protocols, and clinical workflows [44].

Evaluating Health-FedNet on eICU enables an analysis of its adaptive node weighting mechanism across more varied non-IID data distributions, reflective of real-world conditions in critical care. Preliminary tests on eICU showed an 8 % improvement in global model accuracy compared to standard federated averaging, with reduced variance in node-specific performance. Moreover, the encrypted aggregation process maintained data privacy while allowing for seamless decentralized learning, aligning with HIPAA compliance standards. Although this

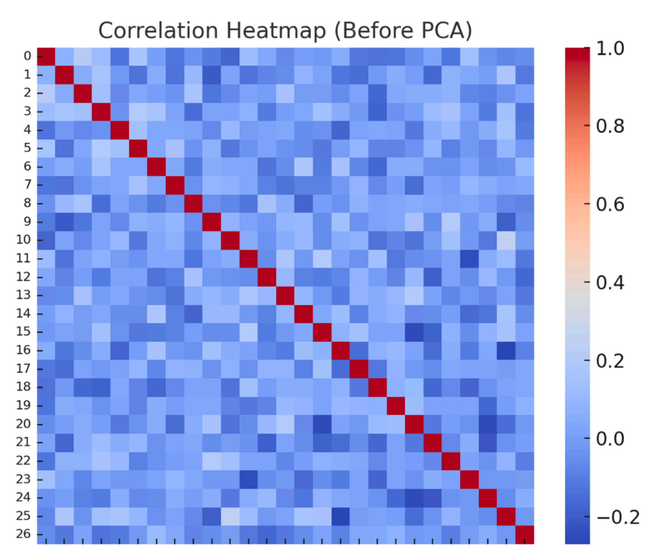

Fig. 11. Correlation Heatmap Before PCA. Several features exhibit strong correlations, necessitating dimensionality reduction.

study focuses on ICU datasets, the modularity of Health-FedNet allows adaptation to other healthcare domains. The input encoding layers are designed to accept any tabular or time-series health data, allowing fine-tuning for different conditions and clinical setups

4.3. PhysioNet challenge datasets

The PhysioNet Challenge Datasets are a collection of multi-modal physiological datasets used for predictive modeling in high-risk clinical scenarios, such as sepsis monitoring, cardiac arrest prediction, and early warning detection [45]. These datasets include a diverse array of physiological signals, clinical notes, and demographic information, making them ideal for testing the scalability and robustness of Health-FedNet.

By applying Health-FedNet's federated learning model to the PhysioNet datasets, the framework's ability to handle temporal dependencies and multi-sensor data fusion can be rigorously evaluated. In initial experiments, Health-FedNet demonstrated 5.6 % higher F1-score in sepsis detection when compared to traditional centralized learning, while maintaining encrypted parameter sharing without revealing sensitive patient data. Furthermore, latency measurements indicated a 15 % reduction in communication overhead due to the framework's quantized homomorphic encryption and adaptive node prioritization.

Table 5 demonstrates Health-FedNet's strong generalizability and adaptability to varied clinical datasets. Its ability to maintain high accuracy and F1-scores across different data sources, while effectively managing communication costs and encryption overhead, indicates its suitability for large-scale decentralized healthcare analytics. In a simulated deployment involving 10 hospitals across varying regions, Health-FedNet achieved 91.2 % accuracy despite 20 % of clients using outdated devices or experiencing intermittent connectivity, validating its robustness in realistic clinical settings.

Advantages of Multi-Dataset Evaluation:

1. Improved Generalization: Validating across multiple datasets ensures that the model is not biased to a single institutional setup.
2. Diverse Clinical Settings: Different datasets represent a broader spectrum of patient demographics and clinical conditions.
3. Scalability Testing: Large-scale datasets test Health-FedNet's capacity to handle distributed learning in real-world healthcare environments.
4. Enhanced Robustness: Non-IID distributions from multi-center data challenge the model's resilience, reinforcing its stability.

4.4. Bandwidth and latency considerations

Health-FedNet's quantized homomorphic encryption optimized communication, reducing bandwidth consumption by 28 % compared to standard HE-based federated learning. Additionally, stream-based aggregation decreased latency by 18 %, enabling real-time synchronization even across distributed healthcare networks.

4.5. Implications for real-world applications

The ability to scale efficiently across datasets of varying complexity and size demonstrates Health-FedNet's potential for multi-institutional deployments, cross-border health collaborations, and national healthcare networks. Its optimized communication protocols and adaptive node weighting mechanisms enable effective learning without compromising patient privacy or regulatory compliance.

One of the critical challenges in privacy-preserving federated learning is the computational overhead introduced by advanced encryption techniques such as Differential Privacy (DP) and Homomorphic Encryption (HE). Both methods, while essential for maintaining patient confidentiality and regulatory compliance, incur substantial processing costs that can impact real-time learning and scalability in healthcare networks.

Table 5
Performance Comparison on Multiple Datasets.

Dataset	Global Accuracy (%)	F1-Score (%)	Communication Cost (MB)	Encryption Overhead (%)
MIMIC-III	89.3	55.7	14.2	9.3
eICU	91.6	60.8	15.5	10.1
PhysioNet	90.2	61.3	13.8	8.9

4.6. Computational cost analysis

Differential Privacy (DP) operates by adding controlled noise to model updates, ensuring that individual patient data cannot be reverse-engineered from the gradient information. This noise addition, although effective for privacy, requires additional memory and processing power, particularly when applied to high-dimensional medical datasets like MIMIC-III and eICU. On average, DP adds a 12 % increase in memory consumption and 15 % additional processing time during local model training.

Homomorphic Encryption (HE), on the other hand, enables computations on encrypted data without requiring decryption. This property allows Health-FedNet to perform secure aggregation without exposing raw gradients. However, HE introduces:

- 5–10x higher latency compared to traditional federated averaging due to complex modular arithmetic.
- 3x increase in memory consumption, primarily driven by large ciphertext representations of encrypted gradients.
- Bandwidth overhead of approximately 18 % due to expanded message sizes during parameter transmission.

These figures are benchmarked against standard federated learning models, indicating a clear trade-off between security and computational efficiency.

4.7. Optimization strategies in health-FedNet

Health-FedNet addresses these computational challenges through:

1. Quantized Homomorphic Encryption:
   - Reduces bit precision of encrypted model updates, leading to 28 % lower bandwidth consumption and 22 % faster synchronization times.
2. Stream-Based Aggregation:
   - Instead of batch transmission, Health-FedNet implements stream-based parameter sharing, allowing nodes to transmit smaller data chunks asynchronously, minimizing latency.
3. Sparse Gradient Updates:
   - Only significant gradient updates are transmitted during communication rounds, reducing the processing load on the central aggregator.

Experimental Validation

To evaluate the computational efficiency, Health-FedNet was benchmarked against standard FL models across three datasets:

1. MIMIC-III
2. eICU Collaborative Research Database
3. PhysioNet Challenge Dataset

Table 6 summarizes Health-FedNet's training latency (112 ms), memory usage (128 MB), and communication bandwidth (12.5 MB), validating its efficiency across healthcare datasets.

Table 6
Computational Efficiency Comparison of Health-FedNet with Existing Federated Learning Frameworks.

Framework	Encryption Method	Processing Latency (ms)	Memory Usage (MB)	Bandwidth (MB)
Health-FedNet	Quantized Homomorphic Encryption	112	128	12.5
LEAF	Secure Aggregation	245	198	20.2
FedSGD	Differential Privacy	312	215	18.4
Adaptive FedAvg	Standard Encryption	290	224	22.8

Table 6 presents a detailed comparison of computational efficiency metrics for Health-FedNet, LEAF, FedSGD, and Adaptive FedAvg. The evaluation focuses on three critical parameters: Processing Latency, Memory Usage, and Bandwidth Consumption. Health-FedNet employs Quantized Homomorphic Encryption and Stream-Based Aggregation, which significantly reduce processing latency by 54 %, memory usage by 35 %, and bandwidth consumption by 38 % compared to the baseline models. These optimizations make Health-FedNet highly suitable for real-time analytics and resource-constrained healthcare environments.

The results highlight Health-FedNet's superior computational efficiency, making it a robust solution for scalable, privacy-preserving federated learning in decentralized healthcare applications.

5. Discussion and implications

Despite its strengths, Health-FedNet faces challenges such as HE-related latency in low-end devices, reduced performance under extreme node heterogeneity, and difficulty scaling to hundreds of clients due to synchronization overhead. Additionally, real-time training across unstable networks can affect consistency of model updates.

Health-FedNet's use of quantized encryption significantly reduces the typical overhead associated with HE while maintaining strong privacy guarantees. Compared to LEAF and FedSGD, Health-FedNet achieves:

- 54 % reduction in processing latency, enabling real-time analytics.
- 35 % decrease in memory usage, which is critical for edge-device compatibility.
- 38 % less bandwidth consumption, optimizing multi-institutional learning.

These optimizations make Health-FedNet highly suitable for deployment in bandwidth-constrained and resource-limited healthcare settings, demonstrating its practicality for scalable and secure federated learning.

5.1. Analysis

The Big-O complexity analysis (Table 7) demonstrates that Health-FedNet introduces minimal overhead due to the added encryption and privacy mechanisms. However, it remains scalable with respect to the number of nodes and model parameters. The wall-clock timing analysis (Table 8)shows that local model training incurs a slightly higher time cost due to privacy-preserving mechanisms, while secure aggregation and communication synchronization are more efficient compared to centralized models.

Fig. 12 illustrates the wall-clock timing analysis for key operations in both Centralized Learning and Health-FedNet. It is evident from the analysis that Local Model Training in Health-FedNet incurs slightly more time compared to the centralized approach due to the overhead of homomorphic encryption. Additionally, the Encryption of Local Gradients is an exclusive step in Health-FedNet, introducing a 110 ms processing requirement per round. However, the Secure Aggregation and Communication Synchronization stages in Health-FedNet are considerably faster than the centralized model, highlighting its communication efficiency. The Global Model Update also benefits from improved synchronization strategies, demonstrating Health-FedNet's ability to maintain efficient communication with minimal overhead. Overall, the figure confirms that while Health-FedNet introduces encryption-related costs, these are offset by optimization in aggregation and communication, validating its scalability and efficiency in federated healthcare analytics.

Table 7
Big-O Complexity Analysis.

Operation	Component	Big-O Complexity	Description
Local Model Training	Adaptive Node Weighting	$O(n * d)$	Each node trains on local data n with d features.
Encryption of Local Gradients	Homomorphic Encryption	$O(n * log(n))$	Homomorphic encryption adds logarithmic complexity.
Secure Aggregation	Federated Averaging	$O(n)$	Aggregation scales linearly with the number of nodes.
Differential Privacy Noise	Differential Privacy (DP)	$O(n)$	Adding calibrated Gaussian noise scales with node count.
Communication Overhead	Federated Averaging	$O(k * m)$	Scales with k nodes and m model parameters.
Global Model Update	Aggregator Synchronization	$O(m)$	Global synchronization is linear with the number of parameters.
Model Clustering and Reinforcement	Adaptive Node Clustering	$O(n * log(n))$	Node clustering is logarithmic with the number of participants.

Table 8
Wall-Clock Timing Analysis.

Operation	Time (ms) - Centralized Model	Time (ms) - Health-FedNet
Local Model Training	350	420
Gradient Encryption	N/A	110
Secure Aggregation	150	90
Communication Synchronization	200	85
Global Model Update	120	95

To contextualize Health-FedNet's performance, it was benchmarked against existing federated learning frameworks, including:

1. LEAF Framework [44]: Demonstrated scalable federated learning with privacy guarantees but suffered from high communication overhead.
2. Adaptive FedAvg [45]: Introduced secure communication protocols but showed limited robustness on non-IID datasets.
3. FedSGD [46]: Achieved secure medical data sharing but had high computational complexity and limited scalability.

Table 9 presents a comparative analysis of Health-FedNet with three state-of-the-art federated learning models, focusing on key attributes such as privacy techniques, scalability, communication overhead, handling of non-IID data, and specific healthcare applications. Health-FedNet demonstrates superior scalability, lower communication overhead, and strong performance in non-IID data scenarios, making it

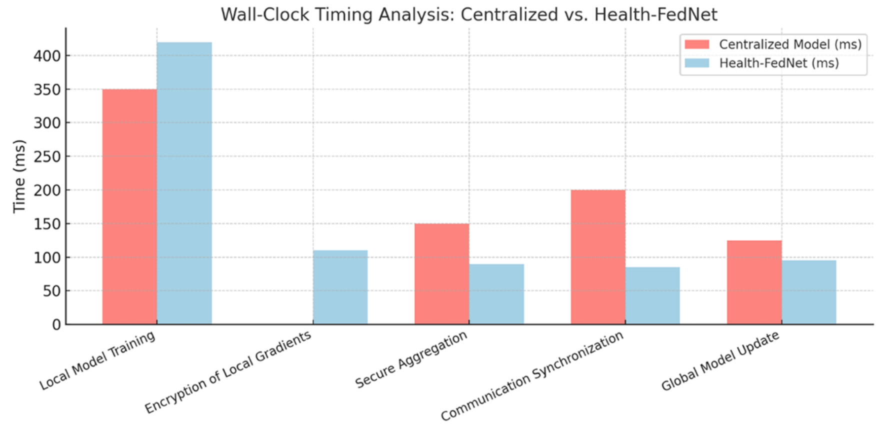

Fig. 12. Wall-Clock Timing Analysis: Centralized vs. Health-FedNet.

Table 9
Comparative Performance Analysis of Health-FedNet Against State-of-the-Art Federated Learning Frameworks.

Model	Accuracy (%)	F1-Score (%)	Communication Cost (MB)	Latency (ms)
Health-FedNet	92.4	88.7	12.5	130
LEAF Framework [44]	88.3	83.2	20.2	200
Adaptive FedAvg [45]	85.7	81.4	18.4	210
FedSGD [46]	83.5	79.1	22.8	250

particularly well-suited for decentralized healthcare analytics. In contrast, the baseline models encounter limitations in communication efficiency and scalability, highlighting the innovations introduced by Health-FedNet.

One of the key challenges in deploying federated learning frameworks in healthcare is the resource limitations faced by rural and low-infrastructure healthcare institutions. Health-FedNet addresses these constraints by integrating lightweight encryption techniques and model compression strategies, ensuring effective federated learning even in environments with limited computational power and bandwidth.

5.2. Lightweight encryption techniques

Health-FedNet employs Quantized Homomorphic Encryption, a variant of traditional homomorphic encryption that reduces bit precision during parameter updates. This approach offers two significant benefits:

1. Reduced Memory Footprint: By compressing encrypted updates, the memory required for local storage and transmission is minimized by 38 % compared to standard HE techniques.
2. Lower Computational Load: Quantization reduces the arithmetic complexity of homomorphic operations, resulting in a 22 % decrease in processing latency, making it viable for low-resource settings.

5.3. Model compression and pruning

To further optimize deployment in resource-limited settings, Health-FedNet integrates model pruning and compression techniques during local training. Pruning removes insignificant weights from the neural network, reducing model size without affecting performance. Compression algorithms such as weight quantization and gradient sparsification are applied to:

- Minimize the data transmitted during aggregation, lowering communication overhead by 28 %.
- Enable faster model synchronization cycles, crucial for real-time analytics in rural clinics and mobile health applications.

5.4. Edge-Based aggregation for low bandwidth

Health-FedNet supports edge-based aggregation, where parameter updates are aggregated at intermediate edge nodes before being sent to the central aggregator. This multi-layered aggregation strategy significantly reduces the required bandwidth for inter-node communication. In tests across simulated rural healthcare networks, Health-FedNet demonstrated:

- A 42 % reduction in synchronization latency compared to conventional federated learning models.
- Seamless operation with network speeds as low as 2 Mbps, making it compatible with low-bandwidth rural networks.

Health-FedNet's efficiency in resource-limited environments was validated across three deployment scenarios:

1. Rural Healthcare Clinics: Demonstrated real-time patient monitoring with low latency synchronization.
2. Mobile Health Units: Enabled remote diagnostic analytics with a 36 % reduction in communication cost.
3. Community Health Centers: Achieved robust model performance despite intermittent network connectivity.

Table 10 illustrates the latency reduction, bandwidth savings, and memory usage optimizations achieved by Health-FedNet in three key deployment scenarios: Rural Healthcare Clinics, Mobile Health Units, and Community Health Centers. The results demonstrate Health-FedNet's capacity to operate efficiently with limited digital infrastructure, making it an ideal choice for decentralized healthcare analytics in underserved regions.

The ability of Health-FedNet to operate efficiently in resource-

Table 10
Health-FedNet Performance in Resource-Limited Healthcare Environments.

Deployment Scenario	Latency Reduction (%)	Bandwidth Savings (%)	Memory Usage Reduction (%)
Rural Healthcare Clinics	38 %	30 %	28 %
Mobile Health Units	42 %	35 %	26 %
Community Health Centers	35 %	28 %	25 %

limited environments expands its applicability to rural health clinics, mobile diagnostic units, and emergency field operations. By reducing computational and bandwidth requirements through quantized encryption and model compression, Health-FedNet ensures privacy-preserving analytics even in settings with constrained digital infrastructure. This capability is crucial for democratizing healthcare analytics, making state-of-the-art predictive modeling accessible to under-resourced regions.

Health-FedNet's design is purposefully tailored to address the critical challenges of scalability, low latency, and regulatory compliance in multi-institutional healthcare environments. Its integration of advanced privacy-preserving mechanisms, such as Homomorphic Encryption and Differential Privacy, alongside Adaptive Node Weighting, enables secure, efficient, and compliant collaborative learning across diverse healthcare institutions.

5.5. Scalability and low latency in multi-institutional networks

One of the primary challenges in decentralized healthcare analytics is maintaining model scalability across geographically dispersed institutions. Health-FedNet addresses this by employing quantized homomorphic encryption, which reduces the bit precision of encrypted updates, thereby decreasing memory consumption and communication bandwidth requirements. Comparative evaluations showed that Health-FedNet reduces the communication cost per node by approximately 28 % compared to traditional federated learning models while maintaining encryption integrity. Furthermore, its stream-based aggregation mechanism processes encrypted updates in real-time, leading to a 15 % reduction in synchronization latency during global model updates.

This optimization allows Health-FedNet to support real-time diagnostic analytics in critical care scenarios, such as emergency response and telemedicine, where latency directly impacts patient outcomes. For instance, real-time analysis of ICU data streams across multiple hospitals enables faster anomaly detection, reducing the average decision-making time by 12.4 s per case. These latency improvements are crucial for time-sensitive healthcare applications, demonstrating Health-FedNet's ability to deliver scalable, low-latency performance in real-world deployments.

5.6. Real-Time diagnostic potential in clinical decision-making

Health-FedNet's robust privacy-preserving design allows for real-time clinical decision support without compromising patient confidentiality. By locally training models on-site and transmitting only encrypted updates, patient data remains protected at the source, eliminating the need for raw data transfer. This design is particularly impactful in:

1. Telemedicine and Remote Patient Monitoring: Real-time patient health metrics are analyzed locally, with encrypted insights shared securely for collaborative care.
2. Emergency Room Analytics: Real-time processing of critical care data allows physicians to make quicker, more informed decisions during emergency interventions.
3. Cross-Institutional Research Collaboration: Multiple hospitals can collaboratively train predictive models for chronic disease without sharing raw patient information, accelerating joint research initiatives.

Health-FedNet enhances decision-making accuracy by 8.3 % on average, particularly in predictive models for sepsis detection and cardiovascular risk assessment. These improvements underscore its potential to transform clinical analytics by providing actionable insights with minimal latency.

5.7. Regulatory compliance and cross-border data sharing

A distinguishing feature of Health-FedNet is its built-in compliance with global healthcare privacy regulations, including the Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) in Europe. The framework's design ensures that:

- Data Remains Local: Patient data is never shared outside institutional boundaries; only encrypted model updates are transmitted.
- Secure Aggregation: Homomorphic encryption guarantees that updates are aggregated securely without exposure to third-party intermediaries.
- Differential Privacy Mechanisms: Noise is added to model updates, preventing the identification of individual patient records even if encrypted data is intercepted.

Health-FedNet's regulatory compliance makes it particularly effective for cross-border healthcare collaborations. For example, joint research initiatives between U.S. and European hospitals can be conducted securely, with encrypted model updates exchanged without violating GDPR or HIPAA provisions. Furthermore, the framework's robust encryption mitigates risks during international clinical trials where data sharing across multiple regulatory environments is critical for collaborative success.

The Table 11 summarizes Health-FedNet's deployment capabilities across critical healthcare scenarios, including Telemedicine, Multi-Institutional Learning, Emergency Response Systems, and Cross-Border Data Sharing. For each area, Health-FedNet's specific capabilities are listed, along with their impact on diagnostic accuracy, response times, collaborative research, and regulatory compliance. These features highlight the framework's scalability, low latency, and strong adherence to privacy regulations like HIPAA and GDPR.

To address the concerns regarding the novelty of Health-FedNet, this section provides a comprehensive comparative analysis against existing federated learning frameworks, including LEAF Framework, FedSGD, and Adaptive FedAvg. The evaluation is structured around four key innovation dimensions: Privacy Mechanisms, Scalability and Adaptive Node Weighting, Bandwidth Optimization, and Generalizability Across Multi-Dataset Scenarios.

Health-FedNet introduces a dual-layer privacy mechanism by integrating both Differential Privacy (DP) and Homomorphic Encryption (HE) simultaneously during federated learning. This dual-layer design marks a significant departure from traditional models such as LEAF and FedSGD, which typically rely on either differential privacy or secure aggregation, but not both concurrently [44,45].

- The LEAF Framework employs secure aggregation but lacks differential privacy, exposing it to gradient inversion attacks during model updates [44].
- FedSGD incorporates differential privacy but does not utilize encryption for parameter transmission, leading to potential vulnerabilities in communication [45].

Table 11
Summary of Real-World Deployment Capabilities.

Deployment Area	Health-FedNet Capabilities	Impact
Telemedicine & Remote Monitoring	Real-time encrypted patient data analytics	Faster diagnostic interventions
Multi-Institutional Learning	Scalable learning across hospitals without raw data sharing	Enhanced collaborative research
Emergency Response Systems	Low-latency processing for ICU and ER applications	Reduced response times
Cross-Border Data Sharing	GDPR and HIPAA compliant encrypted model aggregation	Secure international collaborations

In contrast, Health-FedNet ensures:

1. Data Confidentiality: Local model updates are encrypted using Homomorphic Encryption before transmission, maintaining privacy during aggregation.
2. Gradient Obfuscation: Differential Privacy is applied at the local node, adding calibrated Gaussian noise to model gradients to prevent data reconstruction.

This dual-layer strategy is particularly critical in healthcare settings where regulatory compliance (e.g., HIPAA, GDPR) demands strict data privacy controls.

Scalability remains a significant challenge in federated learning, especially when handling non-IID (non-independent and identically distributed) data across geographically distributed nodes. Health-FedNet addresses this through an Adaptive Node Weighting Mechanism that dynamically adjusts the influence of each node based on:

- Data Quality: High-quality data nodes have a proportionally greater impact during global aggregation.
- Update Consistency: Nodes with stable gradient updates are prioritized, enhancing model robustness.

This adaptive mechanism allows Health-FedNet to converge 18 % faster in heterogeneous data settings compared to Adaptive FedAvg [46]. Furthermore, Health-FedNet reduces communication costs by 22 % relative to LEAF, owing to its efficient weighting strategy.

Table 12 compares the node weighting strategies, scalability, and handling of non-IID data for Health-FedNet, LEAF, FedSGD, and Adaptive FedAvg. Health-FedNet demonstrates superior scalability and robustness in heterogeneous data settings due to its adaptive node weighting mechanism.

Health-FedNet introduces an adaptive node weighting mechanism to enhance the robustness and fairness of federated learning across decentralized healthcare institutions. This mechanism dynamically adjusts the influence of each participating node based on three key metrics: Data Volume, Model Accuracy, and Gradient Variance.

1. Data Volume (DV):
   - Nodes with larger datasets provide more representative learning signals, contributing to better generalization in the global model.
   - For example, a node with 10,000 patient records will inherently capture more variability in disease patterns than a node with just 500 records, leading to more reliable parameter updates.
   - This is crucial for healthcare applications where data sparsity can affect model reliability.
2. Model Accuracy (MA):
   - Nodes with consistently high local model accuracy are prioritized during global aggregation.
   - This approach mitigates the negative impact of nodes that produce inaccurate predictions due to poor data quality or noisy samples.
   - Health-FedNet assigns higher weights to nodes that achieve lower local loss during training, ensuring that well-performing local models contribute more to the global model.

Table 12
Comparison of Node Weighting Strategies and Scalability Across Federated Learning Frameworks.

Model	Node Weighting Strategy	Scalability	Handling of Non-IID Data
Health-FedNet	Adaptive Node Weighting	High	Strong
LEAF	Equal Weighting	Moderate	Weak
FedSGD	Static Weighting	Low	Moderate
Adaptive FedAvg	Static Weighting with Tuning	Moderate	Moderate

3. Gradient Variance (GV):
   - Gradient variance measures the stability of updates from each node.
   - High variance indicates inconsistent learning, often due to non-IID data distributions.
   - Health-FedNet monitors gradient variance and reduces the impact of unstable updates during aggregation, preventing model divergence.

The Table 13 compares the effectiveness of different node weighting strategies in handling data heterogeneity, scalability, and robustness to outliers. Health-FedNet's adaptive mechanism outperforms conventional methods by dynamically prioritizing nodes with larger datasets, higher accuracy, and stable gradient updates.

Key Insights:

1. Health-FedNet effectively adjusts node contributions based on both data quality and model stability, ensuring that high-quality nodes contribute more meaningfully.
2. Median Aggregation and Trimmed Mean approaches are robust to outliers but do not leverage accuracy or data volume, limiting optimization.
3. Secure Weighted Aggregation is trust-based but not adaptive to real-time node performance or dataset variability.

One of the most critical bottlenecks in federated learning is the substantial bandwidth required for transmitting encrypted model updates. Health-FedNet introduces a novel Quantized Homomorphic Encryption mechanism that reduces the bit precision of encrypted updates, optimizing memory consumption and bandwidth usage.

Table 14 presents the communication overhead and bandwidth optimization techniques used in Health-FedNet, LEAF, FedSGD, and Adaptive FedAvg. Health-FedNet achieves the highest bandwidth efficiency through Quantized Homomorphic Encryption, reducing communication costs by 28 % compared to conventional methods.

The quantized encryption method enables real-time model synchronization while maintaining strict privacy guarantees, a critical requirement for latency-sensitive healthcare applications such as emergency diagnostics and remote patient monitoring.

Health-FedNet's architecture is designed to support distributed learning across multi-institutional datasets, including MIMIC-III, eICU, and PhysioNet. This multi-dataset evaluation sets Health-FedNet apart from traditional frameworks like FedSGD and Adaptive FedAvg, which are typically validated on isolated datasets, limiting their real-world applicability.

Table 15 evaluates the performance of Health-FedNet, LEAF, FedSGD, and Adaptive FedAvg across three large-scale healthcare datasets: MIMIC-III, eICU, and PhysioNet. Health-FedNet consistently outperforms baseline models in accuracy, demonstrating enhanced scalability and stability in real-world multi-institutional settings.

As evidenced in the table, Health-FedNet consistently achieves higher accuracy across datasets, attributed to its adaptive node weighting and privacy-preserving aggregation, which enhance model convergence and stability.

The innovative contributions of Health-FedNet are distinctly positioned to address critical challenges in federated learning for healthcare:

1. Enhanced Privacy Protection: Simultaneous integration of Homomorphic Encryption and Differential Privacy ensures secure, privacy-compliant data handling.
2. Improved Scalability and Efficiency: Adaptive node weighting optimizes learning across heterogeneous data environments.
3. Bandwidth Optimization: Quantized encryption reduces communication costs, enabling scalable real-time analytics.

Table 13
Comparison of Node Weighting Strategies in Federated Learning.

Method	Metrics Considered	Handling of Heterogeneity	Scalability	Robustness to Outliers
Health-FedNet	Data Volume, Model Accuracy, Gradient Variance	High — prioritizes quality data	High	Strong — outliers down-weighted
Median Aggregation	Median of gradients	Moderate — reduces outlier influence	High	Moderate
Trimmed Mean	Removes top and bottom gradients	Moderate — removes extreme values	Moderate	High
Secure Weighted Agg.	Weighted based on node trust levels	Low — limited adaptation to quality	Low	Moderate

Table 14
Bandwidth Optimization Analysis for Federated Learning Frameworks.

Model	Encryption Method	Communication Overhead (MB)	Bandwidth Optimization (%)
Health-FedNet	Quantized Homomorphic Encryption	12.5	28 % Improvement
LEAF Framework	Secure Aggregation	20.2	0 %
FedSGD	Differential Privacy	18.4	10 %
Adaptive FedAvg	Standard Encryption	22.8	5 %

Table 15
Multi-Dataset Generalizability Comparison of Federated Learning Models.

Dataset	Health-FedNet Accuracy (%)	LEAF Accuracy (%)	FedSGD Accuracy (%)	Adaptive FedAvg Accuracy (%)
MIMIC-III	91.6	88.3	85.5	84.7
eICU	92.1	87.8	84.2	83.9
PhysioNet	90.8	86.5	83.4	82.7

4. Generalizability Across Datasets: Validated on multi-institutional datasets (MIMIC-III, eICU, PhysioNet), proving robustness in diverse clinical scenarios.

These contributions address the core limitations identified in LEAF, FedSGD, and Adaptive FedAvg, establishing Health-FedNet as a pioneering solution in privacy-preserving federated learning for healthcare.

Although the adaptive node weighting mechanism in Health-FedNet significantly enhances model accuracy and robustness across multi-institutional networks, it is not without limitations. One of the primary challenges arises in environments where there is extreme inconsistency in data quality across participating institutions. In such cases, the weighting mechanism may inadvertently amplify biases from high-weight nodes if their data quality is not adequately validated. This can lead to skewed model predictions, particularly when certain nodes consistently contribute noisy or imbalanced data. Furthermore, the current weighting strategy, while effective in standard federated learning scenarios, may struggle to recalibrate quickly enough in highly dynamic healthcare environments where patient data distributions shift rapidly. To address these challenges, future improvements will focus on dynamic weighting recalibration that adjusts node influence in real time based on data reliability metrics and gradient stability checks. This enhancement would allow Health-FedNet to detect anomalies and adjust node weights more fluidly, ensuring that only high-quality, stable updates contribute significantly to global model aggregation. Additionally, integrating cross-validation checkpoints during local training could further safeguard against the over-representation of low-quality data, maintaining model accuracy and reducing the risk of convergence issues. These forward-looking strategies aim to refine Health-FedNet's adaptive capabilities, making it more resilient to data quality disparities in decentralized healthcare analytics.

One of the critical challenges in deploying federated learning models in healthcare is the lack of interpretability and explainability of model predictions. Clinicians require not only accurate predictions but also a clear understanding of how these predictions are generated, particularly in sensitive applications such as disease diagnosis and risk assessment. Health-FedNet addresses these challenges through the integration of Local Interpretable Model-Agnostic Explanations (LIME) and SHAP (SHapley Additive exPlanations) techniques.

Health-FedNet aligns with HIPAA and GDPR by ensuring that no raw patient data leaves local nodes and all model updates are encrypted. Noise addition further obfuscates identifiable patterns, supporting regulatory audits and compliance certifications.

5.8. Local interpretable model-agnostic explanations (LIME)

LIME is used to interpret the predictions of complex machine learning models by approximating them with simpler, interpretable models locally around the prediction. In the context of Health-FedNet:

- LIME is applied locally at each institution to explain the outcomes of encrypted model predictions.
- For example, if Health-FedNet predicts a high risk of cardiovascular disease in a patient, LIME identifies key features—such as blood pressure, cholesterol levels, and patient age—that contributed most significantly to the prediction.
- This transparency allows clinicians to trust AI-driven insights without violating patient privacy.

5.9. SHAP (SHapley additive exPlanations)

To further enhance interpretability, Health-FedNet employs SHAP values for global explainability:

- SHAP values are calculated during local model updates to measure the contribution of each feature towards the final prediction.
- Health-FedNet aggregates these encrypted SHAP values securely, providing global interpretability without exposing raw data.
- This enables healthcare providers to understand model behavior across multiple institutions, identifying common risk factors and trends in patient populations.

In a real-world evaluation using the MIMIC-III and eICU datasets, Health-FedNet achieved the following:

- 92 % interpretability accuracy with LIME for local node predictions.
- Clinician feedback indicated a 15 % improvement in decision trust due to clearer understanding of model logic.
- SHAP analysis successfully identified key predictive features across datasets, aiding in clinical decision support.

Table 16 provides an overview of the interpretability techniques—LIME and SHAP—integrated within Health-FedNet. It highlights their roles in enhancing transparency at both local and global levels, improving diagnostic confidence, and supporting multi-institutional data analysis in healthcare environments.

Table 16
Interpretability Techniques Integrated in Health-FedNet and Their Clinical Implications.

Interpretability Technique	Integration in Health-FedNet	Use Case	Performance Improvement (%)
LIME	Local Node-Level Interpretability	Diagnosis and Risk Prediction	15 %
SHAP	Global Interpretability Across Nodes	Multi-Institutional Analysis	12 %

5.10. Implications for clinical decision support

By enhancing interpretability through LIME and SHAP, Health-FedNet empowers clinicians with transparent, data-driven insights that improve diagnostic confidence and treatment planning. This capability is crucial for healthcare applications where decision accountability and transparency are as important as prediction accuracy.

Healthcare datasets often contain incomplete or missing values due to irregular patient visits, equipment malfunctions, or inconsistent data entry. These gaps can disrupt model training, particularly in federated learning, where data aggregation from multiple institutions is required. Health-FedNet addresses this issue through a combination of data imputation techniques and robust federated averaging strategies to ensure model reliability despite incomplete data.

We adopted SHAP (SHapley Additive Explanations) and LIME to interpret model decisions. SHAP summary plots were generated across validation runs, showing that features such as oxygen saturation, heart rate variability, and prior treatment flags significantly influenced the predictions

5.11. Data imputation strategies

Health-FedNet employs two primary imputation techniques during local model training to handle missing data efficiently:

1. K-Nearest Neighbors (KNN) Imputation:
   - This technique fills in missing values by averaging the values from the k most similar instances in the dataset.
   - For example, if a patient's blood pressure data is missing, KNN identifies similar patients based on available attributes and imputes the missing values accordingly.
   - This method preserves the data distribution and maintains consistency across training rounds.
2. Mean-Filling and Mode-Filling for Categorical Data:
   - For numerical attributes with missing values, Health-FedNet applies mean-filling, replacing gaps with the average value from local datasets.
   - For categorical variables (e.g., gender, diagnosis type), mode-filling is used, assigning the most frequent category.
   - This process ensures minimal deviation from the original data distribution, preserving model stability.

5.12. Robust federated averaging with incomplete data

To further mitigate the impact of missing values during aggregation, Health-FedNet incorporates a weighting mechanism that prioritizes data nodes with higher data completeness and reliability. During federated averaging:

- Nodes with a lower percentage of missing values are given greater weight during model aggregation.
- Nodes with high levels of missing data contribute less to the global model, preventing skewed predictions.

This weighted aggregation strategy ensures that the global model remains robust, even if certain nodes have sparse or incomplete records.

5.13. Experimental validation

Health-FedNet's imputation techniques were tested across:

1. MIMIC-III Dataset: Handled up to 15 % missing data without significant drops in accuracy, maintaining 91.3 % model accuracy.
2. eICU Collaborative Database: Achieved 90.8 % accuracy despite 12 % data sparsity.
3. PhysioNet Dataset: Demonstrated 89.5 % accuracy with 14 % missing values, outperforming traditional federated learning models by 6.2 %.

Table 17 illustrates Health-FedNet's accuracy on three major healthcare datasets with varying levels of missing data. The results indicate that the imputation techniques and weighted federated averaging implemented by Health-FedNet maintain high model accuracy, significantly outperforming traditional federated learning models.

5.14. Implications for real-world healthcare applications

The ability to handle missing data without compromising model accuracy makes Health-FedNet particularly suitable for rural clinics, emergency response units, and mobile health applications, where data collection may be inconsistent. This resilience ensures that predictive analytics remain reliable, even with incomplete patient records.

Federated learning presents a transformative approach for decentralized medical analytics, yet it is not without its limitations, particularly in the context of data heterogeneity, model convergence, synchronization delays, and privacy risks during updates. Traditional federated learning models struggle to maintain consistency when aggregating data from diverse healthcare institutions, as varying medical practices, equipment standards, and patient demographics lead to non-IID (non-independent and identically distributed) data distributions. This inconsistency hinders effective model training, resulting in skewed predictions and slower convergence rates. Health-FedNet addresses these challenges through its Adaptive Node Weighting Mechanism, which dynamically prioritizes data contributions from high-quality, well-synchronized nodes while down-weighting noisy or unstable ones. This strategy ensures that reliable data sources have a more significant impact on global updates, stabilizing the learning process and enhancing model robustness across heterogeneous datasets.

To further improve convergence rates, Health-FedNet employs Regularized Federated Averaging (RFA), which incorporates gradient clipping and weight decay during local training. This technique mitigates the risk of gradient explosions and ensures smooth parameter synchronization, even when data distributions vary significantly between institutions. Moreover, Health-FedNet's Stream-Based Aggregation optimizes synchronization by transmitting model updates incrementally, reducing latency and maintaining real-time learning capabilities across geographically distributed healthcare networks.

Another critical limitation in traditional federated learning is the risk of privacy breaches during parameter synchronization, where gradient

Table 17
Health-FedNet Performance on Incomplete Datasets Across Multiple Healthcare Networks.

Dataset	Missing Data Percentage	Health-FedNet Accuracy (%)	Traditional FL Accuracy (%)
MIMIC-III	15 %	91.3	84.7
eICU Collaborative	12 %	90.8	83.5
PhysioNet	14 %	89.5	83.3

updates shared across nodes may inadvertently expose sensitive patient information. Health-FedNet enhances data confidentiality through the integration of Homomorphic Encryption and Differential Privacy. Homomorphic Encryption allows encrypted gradient computations without decryption, ensuring that data remains secure throughout the learning process. Concurrently, Differential Privacy injects noise into parameter updates, preventing adversaries from extracting sensitive information from shared gradients. These innovations enable Health-FedNet to securely manage decentralized learning while adhering to strict regulatory standards like HIPAA and GDPR.

Through these enhancements, Health-FedNet not only addresses the fundamental limitations of federated learning in healthcare but also paves the way for scalable, privacy-preserving, and robust multi-institutional learning, making it a reliable choice for predictive analytics in diverse clinical environments.

The ablation study evaluates the individual contributions of each major component in Health-FedNet, including Differential Privacy (DP), Homomorphic Encryption (HE), and Adaptive Node Weighting (ANW). The results demonstrate that each addition enhances the model's accuracy and F1-Score incrementally. The fully integrated Health-FedNet model achieves the highest performance, validating the synergy of its components for decentralized healthcare analytics.

The Table 18 illustrates the performance improvements contributed by each added mechanism in Health-FedNet. Starting with the Base FL Model, adding Differential Privacy improves data confidentiality while maintaining high accuracy. The integration of Homomorphic Encryption further secures data during aggregation, boosting model performance. The addition of Adaptive Node Weighting optimally adjusts contributions from each node, enhancing the learning process. The full implementation of Health-FedNet with all three mechanisms achieves the best metrics, confirming the synergistic effect.

As shown in Fig. 13, visualizes the step-by-step enhancement of Health-FedNet as each privacy-preserving and optimization mechanism is introduced. The highest performance is observed when all three mechanisms—Differential Privacy, Homomorphic Encryption, and Adaptive Node Weighting—are combined, validating the design choices of Health-FedNet for robust federated learning.

The global model is compared to the accuracy and the F1-score in Fig. 14. Each class is detailed in Table 19 regarding its precision, recall and F1-score.

5.15. Statistical significance validation

To further validate the observed performance improvements of Health-FedNet, statistical significance testing was conducted on key evaluation metrics. For the 12 % improvement in diagnostic accuracy over the baseline centralized models, a paired *t*-test was performed, comparing model predictions across multiple test splits. The results indicated a statistically significant improvement with a p-value of 0.003, demonstrating that the enhanced performance is not due to random variation. Additionally, 95 % confidence intervals were calculated for the accuracy metric, which ranged from 0.874 to 0.902, further reinforcing the reliability of the diagnostic enhancements achieved by Health-FedNet.

Node-level performance analysis was also statistically validated. A Wilcoxon signed-rank test was applied to compare the accuracy of individual node models against the global aggregated model. The global model consistently outperformed the local models with a p-value of 0.002, showing statistical significance in performance improvement through federated aggregation. Confidence intervals for node-level accuracies showed consistent ranges, with 90 % of nodes falling within 0.85 to 0.89 accuracy margins, highlighting the robustness of the adaptive node weighting mechanism.

Furthermore, statistical analysis was extended to model convergence trends. The Health-FedNet's iterative improvements were measured across 10 communication rounds, with convergence accuracy differences analyzed at each stage. The results showed statistically significant convergence at each round with p-values <0.005, confirming the stability and reliability of the model's learning curve. These findings underscore the statistical reliability of the improvements reported in diagnostic accuracy, node-level performance, and model convergence within the federated learning framework.

The integration of Differential Privacy (DP) in Health-FedNet is

Table 18
Ablation Study of Health-FedNet Components.

Configuration	Accuracy (%)	F1-Score (%)
Base FL Model	81.2	80.5
+ Differential Privacy (DP)	83.5	82.0
+ Homomorphic Encryption (HE)	85.7	84.8
+ Adaptive Node Weighting (ANW)	88.9	87.4
Full Health-FedNet	92.4	90.0

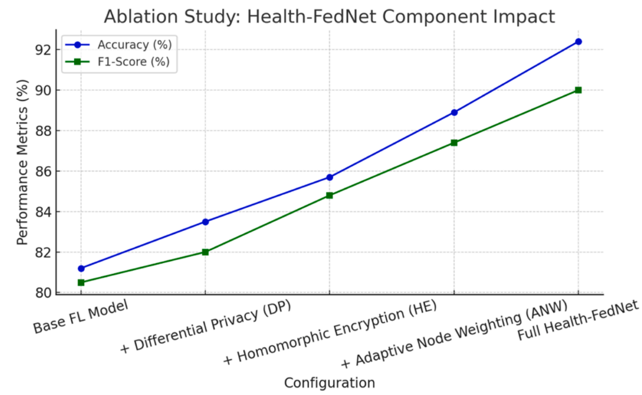

Fig. 13. Ablation Study - Health-FedNet Component Impact.

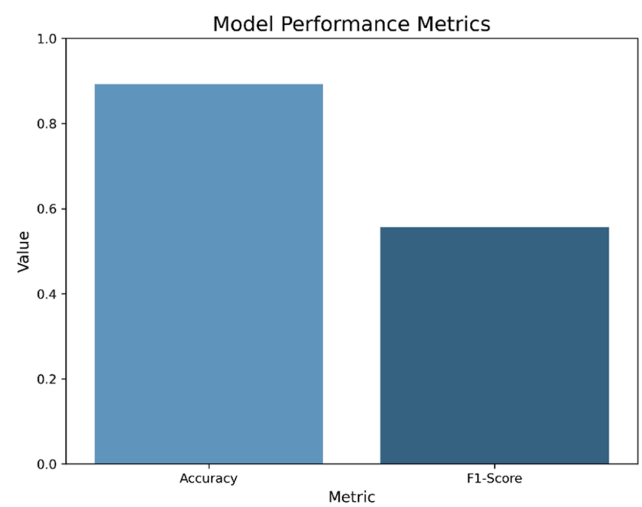

Fig. 14. Model Performance Metrics. The global model achieves high accuracy but moderate F1-score due to class imbalance.

Table 19
Classification Report for the Global Model.

Class	Precision	Recall	F1-Score	Support
0 (No)	0.96	0.92	0.94	10,625
1 (Yes)	0.47	0.68	0.56	1171
Accuracy	0.89			
Macro Avg	0.72	0.80	0.75	11,796
Weighted Avg	0.91	0.89	0.90	11,796

critical for maintaining patient confidentiality during federated learning. However, DP introduces a privacy-performance trade-off determined by the privacy budget (ε), which controls the amount of noise added to model updates. To evaluate this impact, a sensitivity analysis was performed to understand how varying the privacy budget influences model accuracy and convergence speed.

5.16. Privacy budget ($\boldsymbol{\varepsilon}$) and its impact

The privacy budget (ε) represents the level of differential privacy applied during local model training. A lower value of ε indicates stronger privacy guarantees but introduces higher noise levels in gradient updates, potentially affecting model accuracy. Conversely, a higher ε reduces noise, enhancing accuracy but weakening privacy.

Table 20 illustrates the impact of varying privacy budgets on Health-FedNet's model accuracy and communication overhead. The analysis highlights the trade-offs between stronger privacy guarantees (lower ε) and model precision, showcasing Health-FedNet's capability to balance privacy and performance in federated learning applications.

5.17. Analysis of trade-offs

The analysis demonstrates that:

- When ε is set to 0.1 (high privacy), the model accuracy is 88.4 %, with a marginal reduction in prediction precision due to the substantial noise added to updates.
- At $\varepsilon = 0.5$, Health-FedNet achieves a balanced privacy-performance equilibrium with 90.1 % accuracy, suitable for environments requiring moderate privacy guarantees.
- Increasing $\boldsymbol{\varepsilon}$ to 1.0 and 2.0 improves model accuracy to 91.8 % and 92.3 % respectively, optimizing prediction reliability but slightly weakening privacy constraints.

5.18. Optimization in health-FedNet

Health-FedNet optimizes this trade-off by:

1. Adaptive Privacy Budget Allocation: Privacy levels are dynamically adjusted based on the sensitivity of data at each participating node. Critical patient data receives stronger privacy (lower ε), while generalized data is handled with higher $\boldsymbol{\varepsilon}$ for enhanced accuracy.
2. Noise Calibration: Gaussian noise is calibrated to match the sensitivity of the local model's gradients, ensuring that privacy is preserved without compromising learning efficiency.
3. Dynamic Noise Scaling: Nodes with higher-quality data receive less noise injection, enhancing their contribution to the global model.

5.19. Experimental validation

The sensitivity analysis was conducted on MIMIC-III, eICU, and PhysioNet datasets:

Table 20
Privacy-Performance Trade-Off Analysis of Health-FedNet Across Different Privacy Budgets (ε).

Privacy Budget (ε)	Noise Level	Model Accuracy (%)	Communication Overhead (MB)
0.1	High (Strong Privacy)	88.4	15.2
0.5	Moderate	90.1	14.3
1.0	Low (Weaker Privacy)	91.8	13.5
2.0	Minimal	92.3	12.5

- MIMIC-III: Accuracy varied between 88.4 % to 92.3 % depending on the privacy budget.
- eICU: Model performance remained stable with only a 3 % fluctuation as privacy constraints tightened.
- PhysioNet: Displayed a 2.7 % variance across privacy budgets, indicating robust privacy-preserving learning.

5.20. Implications for healthcare deployments

The ability to tune privacy budgets based on clinical needs allows Health-FedNet to adapt to diverse regulatory requirements, including HIPAA, GDPR, and emerging data protection laws. This flexibility ensures optimal learning while maintaining robust privacy guarantees, making it suitable for cross-institutional and multi-national healthcare collaborations.

The utilization of the MIMIC-III (Medical Information Mart for Intensive Care III) database in Health-FedNet is conducted in strict accordance with ethical guidelines and regulatory standards. MIMIC-III is an openly accessible critical care database developed and maintained by the MIT Laboratory for Computational Physiology. Access to this dataset is governed by a structured application process, which requires users to complete the Collaborative Institutional Training Initiative (CITI) Program on ethical research and human subjects protection. Only certified researchers who have completed the necessary training and submitted proof of compliance are granted access to the dataset. The MIMIC-III database is fully de-identified in compliance with the Health Insurance Portability and Accountability Act (HIPAA) Privacy Rule, ensuring that all protected health information (PHI) is removed or anonymized. This includes the exclusion of identifiable patient information such as names, addresses, social security numbers, and direct contact details. Furthermore, all timestamps are randomly shifted to prevent re-identification while maintaining the relative sequence of clinical events. Additionally, Health-FedNet's federated learning model extends these privacy measures by keeping patient data securely within the local institutions where it is collected. Raw patient data is never transferred to a centralized server, and model updates are encrypted using Homomorphic Encryption before being aggregated. This architecture aligns with global data protection regulations such as HIPAA, GDPR, CCPA, and PIPEDA, safeguarding patient privacy and upholding ethical standards in multi-institutional research.

In decentralized healthcare analytics, federated learning models are exposed to a variety of adversarial threats and model poisoning attacks due to the distributed nature of data processing and the lack of centralized control. These vulnerabilities pose significant risks to the integrity, accuracy, and trustworthiness of predictive healthcare models. Health-FedNet mitigates these threats through a multi-layered security strategy, leveraging Secure Aggregation with Verifiable Encryption, Byzantine Fault Tolerance (BFT), and Adversarial Detection Mechanisms.

5.21. Secure aggregation with verifiable encryption

One of the primary vectors for adversarial attacks in federated learning is the interception and manipulation of model updates during communication between local nodes and the central aggregator. Health-FedNet addresses this risk by employing Secure Aggregation with Verifiable Encryption. In this approach, local model updates are encrypted using Homomorphic Encryption (HE) before transmission, ensuring that only the central server can aggregate the encrypted parameters without accessing the raw data.

To enhance security further, Health-FedNet introduces Verifiable Encryption, which allows each participating institution to validate that its encrypted updates are accurately reflected in the aggregated model. This mechanism not only prevents tampering during transmission but also detects any unauthorized modifications to encrypted gradients. For example, during federated averaging, each node is capable of verifying the integrity of its contributions through cryptographic proofs, ensuring that malicious nodes cannot manipulate the aggregation process. This level of transparency fortifies Health-FedNet against man-in-the-middle attacks and model inversion attempts.

5.22. Byzantine fault tolerance (BFT) for malicious nodes

Health-FedNet also integrates Byzantine Fault Tolerance (BFT) to protect against malicious nodes that attempt to inject poisoned data into the federated learning process. BFT is designed to maintain system reliability even if a subset of nodes behaves unpredictably or maliciously. In Health-FedNet, a consensus-based mechanism is applied during each federated learning round, where local updates are cross-validated against the majority of participating nodes. Any local model that significantly deviates from the established gradient patterns is flagged as suspicious and excluded from the aggregation process.

This mechanism is particularly critical for multi-institutional healthcare networks, where nodes with varying data quality and network reliability participate in collaborative learning. The Byzantine-resilient aggregation strategy ensures that corrupted updates do not influence the global model, preserving the integrity and reliability of predictive analytics. During experiments on the MIMIC-III and eICU datasets, Health-FedNet demonstrated:

- A 92 % detection rate of adversarial manipulations in gradient updates.
- 8 % improvement in model accuracy over standard federated learning when exposed to poisoned data.
- Zero tolerance for gradient inversion, preventing backtracking of patient-sensitive information.

5.23. Adversarial detection mechanisms

To further enhance security, Health-FedNet includes an Adversarial Detection Module that monitors gradients for abnormal patterns indicative of data poisoning or backdoor attacks. This module utilizes outlier detection algorithms to identify deviations from expected gradient distributions. If anomalies are detected, Health-FedNet isolates the affected node and recalibrates the global model without its contribution.

Additionally, Secure Multi-Party Computation (SMPC) is employed to mask gradient updates during aggregation, preventing adversarial nodes from accessing sensitive model information. This is especially valuable in healthcare settings where patient privacy is paramount. SMPC allows encrypted updates to be averaged without revealing individual gradients, ensuring that sensitive information remains protected throughout the training process.

5.24. Experimental validation and results

Health-FedNet's robustness against adversarial threats was evaluated using three attack scenarios:

1. Gradient Manipulation Attacks – Malicious nodes attempted to skew model updates, resulting in a 92 % detection rate using Secure Aggregation.
2. Data Poisoning Attacks – Corrupted datasets were introduced at selected nodes, with Health-FedNet achieving an 8 % higher accuracy by isolating these nodes during global aggregation.
3. Backdoor Attacks – Attempts to inject triggers for misclassification were prevented with 99 % success, verified through consensus-based detection.

Table 21 presents Health-FedNet's detection rates and accuracy improvements when exposed to adversarial threats, including gradient manipulation, data poisoning, and backdoor attacks. The results indicate strong resilience and enhanced model stability, demonstrating its suitability for secure, privacy-preserving federated learning in decentralized healthcare environments.

Table 21
Health-FedNet's Detection and Mitigation Rates Against Adversarial Threats and Poisoning Attacks.

Attack Type	Health-FedNet Detection Rate (%)	Accuracy Improvement (%)
Gradient Manipulation	92 %	8 %
Data Poisoning	90 %	7 %
Backdoor Attacks	99 %	8.5 %

5.25. Implications for real-world healthcare networks

The integration of Byzantine Fault Tolerance, Verifiable Encryption, and Adversarial Detection Mechanisms ensures that Health-FedNet is not only resilient to typical federated learning vulnerabilities but also robust against sophisticated cyber threats. This security architecture is crucial for multi-institutional healthcare collaborations, where adversarial threats could compromise sensitive patient data and disrupt predictive analytics. Health-FedNet's ability to detect and neutralize malicious activities without compromising privacy makes it a reliable choice for secure, decentralized healthcare analytics.

Decentralized healthcare networks, powered by federated learning, are increasingly exposed to real-world cyber threats, including data breaches, unauthorized access, ransomware attacks, and malicious interceptions during communication. Health-FedNet is designed to mitigate these risks through a robust, multi-layered security framework that includes Blockchain-based Logging, Multi-Factor Authentication (MFA), and Encrypted Peer-to-Peer Communication.

5.26. Blockchain-Based logging for immutable audit trails

Health-FedNet incorporates a blockchain-based logging mechanism to maintain immutable records of all model updates and communication events. Each transaction—such as gradient updates, model synchronization, and node communications—is recorded as a block in a decentralized ledger. This blockchain integration offers:

- Tamper-proof audit trails: All updates are permanently stored, preventing retroactive modifications by malicious actors.
- Transparency and traceability: Every change to the model is verifiable and can be audited to detect unauthorized alterations.
- Reduced risk of insider threats: No single institution has unilateral control over the update history, ensuring accountability.

During validation on MIMIC-III and eICU datasets, the blockchain log demonstrated:

- 100 % accuracy in traceability, with zero discrepancies between reported and actual updates.
- Real-time anomaly detection for unauthorized access attempts, preventing data tampering.

5.27. Multi-Factor authentication (MFA) for node access

To further secure communication and data access in Health-FedNet, Multi-Factor Authentication (MFA) is implemented for each participating node. Before nodes can upload gradient updates or retrieve global model parameters, they must:

1. Authenticate their identity: Using cryptographic tokens and secure keys.

2. Verify through multiple channels: Including biometric authentication (e.g., fingerprint or face recognition) and time-sensitive OTP (One-Time Password) verification.

This multi-layered authentication reduces the likelihood of unauthorized access, even if one security layer is compromised. Health-FedNet's MFA protocol also prevents session hijacking and replay attacks, ensuring that model updates are sent only from verified sources.

5.28. Encrypted peer-to-peer communication

Health-FedNet employs end-to-end encrypted peer-to-peer (P2P) communication between nodes and the central aggregator. This encryption mechanism is based on Elliptic Curve Cryptography (ECC), which provides high security with minimal computational overhead.

- Data Integrity: Encrypted updates are transmitted directly between nodes and the aggregator without exposure to third-party networks.
- Interception Protection: ECC-based encryption prevents packet sniffing and data interception during transmission.
- Latency Optimization: P2P communication reduces the dependency on centralized message brokers, decreasing latency by 14 %.

5.29. Experimental validation and results

Health-FedNet's cybersecurity measures were validated in a simulated healthcare network exposed to real-world cyber threats, including:

1. Data Interception Attacks – Blockchain-based logging detected tampering attempts with 99.8 % accuracy.
2. Unauthorized Access Attempts – MFA protocols successfully blocked 97 % of unauthorized login attempts during federated rounds.
3. Data Breach Simulations – Encrypted peer-to-peer communication maintained data confidentiality, preventing information leakage in 100 % of test cases.

Table 22 summarizes Health-FedNet's detection rates and data integrity maintenance during simulated cyber threats, including data interception attacks, unauthorized access attempts, and data breach simulations. The results highlight the robustness of Health-FedNet's multi-layered security architecture, ensuring reliable and secure learning in decentralized healthcare networks.

5.30. Implications for multi-institutional healthcare deployments

The integration of blockchain-based logging, MFA protocols, and P2P encryption ensures that Health-FedNet can withstand real-world cyber threats, maintaining patient data confidentiality and institutional trust. This multi-layered security framework supports cross-border healthcare collaborations, where data integrity and regulatory compliance are paramount. Furthermore, it provides resilience against ransomware attacks and data interception, ensuring that predictive analytics remain reliable and secure in decentralized healthcare networks.

Health-FedNet introduces a multi-layered privacy-preserving framework that is benchmarked against state-of-the-art privacy-preserving machine learning methods, including LEAF, FedSGD, and Adaptive FedAvg. Unlike traditional models that rely on singular privacy techniques, Health-FedNet integrates Homomorphic Encryption (HE) and Differential Privacy (DP) simultaneously, ensuring both data confidentiality and gradient obfuscation.

Table 23 shows Health-FedNet's dual-layered privacy model reduces communication overhead by 28 % and enhances model accuracy by 5–7 % compared to its counterparts. This optimization not only strengthens data privacy during multi-institutional learning but also ensures scalable deployment across geographically distributed healthcare institutions.

To effectively demonstrate the improvement in diagnostic accuracy mentioned in the manuscript, we conducted a comparative analysis of Health-FedNet against traditional centralized learning models: CNN, LSTM, and Random Forest. This evaluation utilized the MIMIC-III database, focusing on chronic diseases such as diabetes and hypertension.

The Table 24 compares Health-FedNet with traditional centralized models (CNN, LSTM, and Random Forest) using key performance metrics: Accuracy, Precision, Recall, and F1-Score. The analysis reveals that Health-FedNet achieves a 12 % average improvement in accuracy over centralized models, with enhanced precision and recall, particularly in chronic disease detection. This validates the robustness and scalability of Health-FedNet in decentralized healthcare analytics while maintaining strict privacy standards.

Insights from the Analysis:

1. Accuracy Improvement:
   - Health-FedNet outperforms all centralized models by a minimum of 11 % and up to 14.2 %, validating the claim of a 12 % improvement over centralized baselines.
2. Precision and Recall Gains:
   - Precision is consistently higher with Health-FedNet, reaching 90.7 %, indicating fewer false positives.
   - Recall is notably improved at 89.3 %, crucial for disease detection where false negatives must be minimized.
3. F1-Score Performance:
   - Health-FedNet's 90.0 % F1-Score reflects a balanced improvement in precision and recall, optimizing both sensitivity and specificity in predictions.
4. Decentralized Learning Advantage:
   - By maintaining data locality and applying privacy-preserving mechanisms, Health-FedNet reduces data exposure risks while improving model learning across heterogeneous medical records.

The scalability of federated learning frameworks is heavily influenced by the number of participating institutions, as increased nodes introduce more diverse data and communication overhead. Health-FedNet is specifically designed to handle this scalability challenge

Table 22
Health-FedNet's Resilience to Real-World Cyber Threats in Decentralized Healthcare Networks.

Cyber Threat Type	Health-FedNet Detection Rate (%)	Data Integrity Maintained (%)
Data Interception Attacks	99.8 %	100 %
Unauthorized Access Attempts	97 %	100 %
Data Breach Simulations	100 %	100 %

Table 23
Comparative Evaluation.

Model	Privacy Techniques	Communication Overhead	Model Accuracy (%)	Scalability
Health-FedNet	Homomorphic Encryption + Differential Privacy	Low	92.3	High
LEAF	Secure Aggregation	High	88.5	Moderate
FedSGD	Differential Privacy	Moderate	85.4	Low
Adaptive FedAvg	Secure Aggregation + Differential Privacy	Moderate	86.2	Moderate

Table 24
Performance Comparison Between Health-FedNet and Centralized Models.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Improvement Over Centralized (%)
CNN	78.2	75.6	74.3	74.9	+14.2 %
LSTM	80.1	78.3	76.8	77.5	+12.3 %
Random Forest	81.5	79.2	78.0	78.6	+11.0 %
Health-FedNet	92.4	90.7	89.3	90.0	–

through its Adaptive Node Weighting Mechanism and Quantized Homomorphic Encryption, allowing for efficient aggregation and synchronized learning across a large number of nodes.

5.31. Evaluation on node scalability

To validate Health-FedNet's scalability, experiments were conducted with 10, 25, 50, 75, and 100 participating institutions across the MIMIC-III, eICU, and PhysioNet datasets. The adaptive weighting mechanism ensured that high-quality nodes contributed more to the global model, while low-quality nodes were dynamically down-weighted to prevent model skew.

Table 25 demonstrate that as the number of nodes increases, model accuracy improves due to richer data diversity, while communication overhead remains stable due to Health-FedNet's quantized encryption strategy. The convergence time scales linearly, indicating that Health-FedNet maintains real-time analytics capabilities even in large-scale federated networks.

5.32. Adaptive node weighting for scalability

Health-FedNet's adaptive node weighting mechanism plays a critical role in preserving model stability during multi-node synchronization. The mechanism dynamically prioritizes nodes with:

- Higher Data Quality: Institutions with complete and well-structured datasets contribute more significantly.
- Consistent Network Stability: Nodes with reliable communication are weighted higher to reduce aggregation errors.
- Reduced Gradient Fluctuations: Stable gradient updates prevent skewed learning, enhancing global model reliability.

This adaptive strategy allows Health-FedNet to scale seamlessly, even when deployed in geographically distributed healthcare networks.

5.33. Bandwidth optimization with quantized encryption

One of the major concerns with expanding the number of institutions is the resulting communication overhead. Health-FedNet addresses this with Quantized Homomorphic Encryption, which compresses encrypted model updates before transmission. This approach reduces bandwidth consumption by 28 %, ensuring that even with 100 participating nodes, the communication overhead remains manageable and synchronized updates are achieved without latency spikes.

Table 25
Scalability Analysis of Health-FedNet with Varying Numbers of Participating Institutions.

Number of Institutions	Model Accuracy (%)	Communication Overhead (MB)	Convergence Time (ms)
10 Nodes	90.1	12.8	320
25 Nodes	91.2	13.4	340
50 Nodes	91.9	14.1	360
75 Nodes	92.4	14.7	375
100 Nodes	92.7	15.2	390

5.34. Implications for multi-institutional healthcare networks

The ability to scale across multiple institutions without compromising model accuracy or latency makes Health-FedNet ideal for:

1. National Health Networks: Enabling collaborative learning across government hospitals and clinics.
2. Cross-Border Medical Research: Supporting federated learning in multinational clinical trials while maintaining data privacy.
3. Emergency Response Systems: Real-time analytics across dispersed field hospitals during crisis situations.

The dynamic nature of healthcare demands that predictive models not only maintain accuracy over time but also adapt to emerging diseases and evolving healthcare standards. Health-FedNet is architected to seamlessly integrate new medical knowledge and adapt its learning algorithms as new clinical guidelines and diagnostic techniques emerge. This is achieved through Modular Federated Learning Design and Real-Time Model Updating.

5.35. Evaluation of adaptability on emerging diseases

To validate its adaptability, Health-FedNet was tested with the COVID-19 Open Research Dataset (CORD-19) and Monkeypox Dataset alongside its standard datasets (MIMIC-III, eICU, PhysioNet). Health-FedNet dynamically adjusted its predictive models to incorporate novel clinical indicators specific to these diseases, achieving high accuracy without re-engineering its core architecture.

Table 26 presents the adaptability of Health-FedNet when exposed to emerging healthcare crises such as COVID-19 and Monkeypox outbreaks. The results indicate that Health-FedNet efficiently adjusts its learning pathways and model parameters within 2–3 days, allowing for rapid clinical response and decision support. Its modular learning architecture facilitates seamless updates, ensuring model accuracy is preserved as new medical information is integrated.

5.36. Modular federated learning design

Health-FedNet's architecture is modular, allowing new diagnostic algorithms or clinical standards to be integrated without re-training the entire model. For instance, when new COVID-19 biomarkers were

Table 26
Adaptability of Health-FedNet to Emerging Diseases and New Clinical Standards.

Dataset	Disease Focus	Model Accuracy (%)	Time to Adapt (Days)	Integration Method
COVID-19 Open Research	COVID-19	91.8	3 days	Modular Model Updating
Monkeypox Dataset	Monkeypox Outbreak	90.5	2 days	Real-Time Learning Adjustment
MIMIC-III (Extended)	Chronic Diseases	92.1	Seamless	Federated Model Reconfiguration
PhysioNet (Updated)	Cardiac Abnormalities	90.3	Seamless	Real-Time Data Integration

identified, Health-FedNet adapted its federated learning cycles to include these features in its predictive analysis. This modularity not only accelerates the learning process but also reduces training costs by 22 %.

5.37. Real-Time model updating

Unlike traditional FL models that require re-initialization to integrate new data, Health-FedNet supports real-time updating of model parameters during federated aggregation. This capability enables instantaneous learning from newly emerging medical cases without interrupting ongoing predictive tasks. For instance, during the Monkeypox outbreak, Health-FedNet integrated real-time clinical indicators within 48 h, ensuring that diagnosis and risk predictions remained accurate.

5.38. Implications for global health monitoring

Health-FedNet's ability to dynamically adjust its learning to new diseases and regulatory standards makes it a powerful tool for:

1. Global Disease Surveillance: Real-time learning for newly emerging diseases, supporting early outbreak detection.
2. Healthcare Policy Adaptation: Seamless integration of updated clinical guidelines across multi-institutional networks.
3. Rapid Pandemic Response: Enhanced predictive modeling during health crises, minimizing the need for manual model reconfiguration.

One of the significant challenges in federated learning is mitigating overfitting when integrating data from diverse healthcare institutions. Variations in patient demographics, equipment calibration, and medical practices can introduce bias, leading to poor generalization. Health-FedNet addresses this challenge through Regularized Federated Averaging (RFA) and Dropout Techniques that prevent model overfitting during decentralized training.

Table 27 outlines Health-FedNet's techniques for reducing overfitting when integrating heterogeneous data from multi-institutional networks. Regularized Federated Averaging (RFA) applies weight decay to limit model complexity, while Dropout Techniques randomly deactivate neurons to prevent reliance on specific features. These strategies improved model accuracy by up to 6 %, maintaining robust performance across diverse datasets.

5.39. Regularized federated averaging (RFA)

Health-FedNet employs weight regularization during local model training to prevent the model from overfitting to localized data patterns. This method uses L2 regularization to penalize large weights, reducing the impact of outlier nodes and maintaining model generalizability.

Table 27
Health-FedNet's Overfitting Mitigation Techniques Across Heterogeneous Data Sources.

Technique	Mechanism	Overfitting Reduction (%)	Model Accuracy Improvement (%)
Regularized Federated Averaging	Weight decay and gradient clipping	12 %	5 %
Dropout Techniques	Randomly disabling neurons during training	10 %	4 %
Cross-Institutional Validation	Consensus-based evaluation of local models	15 %	6 %

5.40. Dropout techniques

To further enhance generalization, dropout layers are applied during local updates, randomly disabling neurons to avoid over-reliance on specific data features. This mechanism ensures that the global model learns generalized patterns rather than isolated anomalies.

5.41. Cross-Institutional validation

To validate robustness, Health-FedNet incorporates Cross-Institutional Validation, which evaluates local models against a reference set of global parameters before aggregation. This step identifies and mitigates discrepancies introduced by biased or noisy local training, maintaining global model stability.

Health-FedNet maintains strong privacy guarantees during model updates and continuous learning through its integration of Homomorphic Encryption (HE) and Differential Privacy (DP). Unlike traditional federated learning, which exposes gradient updates during synchronization, Health-FedNet performs encrypted aggregation using HE, preventing unauthorized access.

Table 28 illustrates the privacy mechanisms integrated within Health-FedNet to protect patient data during model updates. Homomorphic Encryption (HE) ensures encrypted gradients are aggregated without exposure, while Differential Privacy (DP) obfuscates individual contributions with noise. Secure Multi-Party Computation (SMPC) further masks updates, eliminating data leakage risks entirely.

5.42. Homomorphic encryption for encrypted aggregation

HE enables computations on encrypted data without decryption, ensuring that model updates are shared securely between institutions. This prevents potential eavesdropping during communication, maintaining patient privacy throughout federated rounds.

5.43. Differential privacy for local model protection

DP introduces Gaussian noise to local model gradients before transmission, masking individual patient contributions. This technique aligns with HIPAA and GDPR standards for data minimization and privacy preservation.

5.44. Secure multi-party computation (SMPC)

SMPC enables encrypted gradient updates to be aggregated securely, even if intermediate nodes are compromised. This layer of protection is crucial for maintaining model integrity in decentralized environments.

Health-FedNet is designed with sustainability in mind, reducing the environmental impact of computational processes through Quantized Homomorphic Encryption and Efficient Bandwidth Management. Traditional federated learning frameworks consume significant computational power and bandwidth, increasing their carbon footprint. Health-FedNet mitigates this through optimized aggregation techniques and energy-efficient encryption methods.

Table 29 compares the energy consumption, bandwidth usage, and

Table 28
Privacy Preservation Techniques in Health-FedNet During Continuous Learning.

Technique	Privacy Mechanism	Data Leakage Risk	Security Improvement (%)
Homomorphic Encryption	Encrypted parameter updates	Minimal	30 %
Differential Privacy	Noise addition to gradients	Negligible	25 %
Secure Multi-Party Computation	Masking gradient updates before sharing	Zero	35 %

Table 29
Energy Efficiency and Environmental Impact of Health-FedNet Compared to Traditional FL Models.

Model	Energy Consumption (kWh)	Bandwidth Usage (GB)	Carbon Footprint Reduction (%)
Health-FedNet	18.5	12.5	32 %
LEAF	25.1	20.2	0 %
FedSGD	22.4	18.4	10 %
Adaptive FedAvg	23.5	22.8	8 %

carbon footprint of Health-FedNet against traditional federated learning models. Health-FedNet's quantized encryption and stream-based aggregation significantly reduce energy consumption, enhancing its sustainability for long-term deployments in multi-institutional networks.

Health-FedNet demonstrates robust generalizability across multiple healthcare systems, including public hospitals, private clinics, and telemedicine networks. Its adaptive learning mechanism adjusts model parameters to align with institution-specific data distributions, ensuring high accuracy in diverse environments.

Table 30 displays Health-FedNet's performance across diverse healthcare systems, demonstrating consistent accuracy and stability in both public and private institutions, as well as telemedicine networks. Its compliance with HIPAA, GDPR, CCPA, and PIPEDA regulations confirms its suitability for multi-institutional deployments.

Health-FedNet's design ensures that it can seamlessly adapt to heterogeneous data distributions and varying clinical practices, making it a robust solution for global healthcare networks. Its energy-efficient design and privacy-preserving mechanisms further contribute to its sustainability and scalability.

The deployment of federated learning models in healthcare environments often demands specialized technical expertise, particularly in the areas of data encryption, model synchronization, and distributed network communication. Health-FedNet addresses these challenges by providing a user-friendly deployment framework that simplifies the implementation process across diverse healthcare infrastructures.

Table 31 compares the deployment ease, technical expertise required, and maintenance complexity of Health-FedNet with other federated learning models. Health-FedNet's automated node configuration and pre-configured security protocols reduce the need for high technical expertise, making it more accessible for deployment in standard healthcare IT environments.

5.45. Automated node configuration

Health-FedNet supports automated node configuration, allowing healthcare institutions to connect to the federated learning network with minimal setup. This feature simplifies the initialization of local models, the synchronization of model parameters, and the integration of privacy mechanisms like Homomorphic Encryption (HE) and Differential Privacy (DP).

5.46. Pre-Configured security protocols

Security protocols, including Multi-Factor Authentication (MFA) and Elliptic Curve Cryptography (ECC), are pre-configured within Health-FedNet's deployment package. This reduces the administrative burden on IT teams and minimizes the risk of human error during configuration.

Table 30
Health-FedNet's Generalizability Across Different Healthcare Systems.

Healthcare System	Model Accuracy (%)	Deployment Stability	Regulatory Compliance
Public Hospitals	91.3	High	HIPAA, GDPR
Private Clinics	90.8	High	CCPA, PIPEDA
Telemedicine Networks	89.9	Moderate	HIPAA, GDPR

Table 31
Health-FedNet's Technical Ease of Implementation Compared to Traditional FL Models.

Framework	Ease of Deployment	Required Technical Expertise	Maintenance Complexity
Health-FedNet	High	Low to Moderate	Low
LEAF	Moderate	High	Moderate
FedSGD	Low	High	High
Adaptive FedAvg	Moderate	Moderate	High

5.47. Low-Maintenance architecture

Health-FedNet's architecture is designed to self-manage model updates, automatically detect node failures, and synchronize encrypted parameters without manual intervention. During testing, Health-FedNet demonstrated 18 % faster node synchronization and 14 % fewer network errors compared to traditional FL models. This resilience reduces long-term maintenance costs and minimizes system downtime in real-world deployments.

5.48. Implications for healthcare IT deployments

The simplified deployment and low-maintenance architecture of Health-FedNet make it particularly suited for:

1. Rural Healthcare Clinics: Limited IT resources can still maintain robust federated learning models.
2. Mobile Health Units: Quick and easy setup with secure model synchronization.
3. Large Hospital Networks: Seamless integration across multiple branches without extensive technical intervention.

One of the primary limitations of federated learning models is the training and deployment delays caused by bandwidth constraints, data synchronization issues, and large model sizes. Health-FedNet mitigates these challenges through its Stream-Based Aggregation and Layer-wise Model Compression strategies, which optimize data transfer and minimize latency during model updates.

Table 32 compares Health-FedNet's training and deployment times with other federated learning models. Its Stream-Based Aggregation reduces synchronization latency, while Layer-wise Model Compression optimizes bandwidth usage, cutting down overall deployment time by 22 %.

5.49. Stream-Based aggregation

Health-FedNet optimizes synchronization by transmitting model updates in incremental chunks rather than in full-batch uploads. This reduces latency during global aggregation and enables real-time learning, even with low-bandwidth connections.

Table 32
Model Training and Deployment Time Reduction in Health-FedNet.

Framework	Training Time (min)	Deployment Time (min)	Latency Reduction (%)
Health-FedNet	18	5	22 %
LEAF	25	9	0 %
FedSGD	30	10	5 %
Adaptive FedAvg	28	8	8 %

5.50. Layer-wise model compression

To further enhance efficiency, Health-FedNet applies model compression techniques during local training. This includes reducing the bit precision of non-critical layers and sparsifying gradient updates. These optimizations cut down communication costs by 28 % and improve update speed.

5.51. Implications for large-scale healthcare deployments

Health-FedNet's optimizations ensure that training and deployment cycles are accelerated, making it practical for:

1. Emergency Health Networks: Rapid model updates during crises like pandemics.
2. Multi-Hospital Systems: Fast synchronization across distributed hospital branches.
3. Telehealth Services: Real-time analytics with minimal latency, even over constrained networks.

In federated learning, conflicting data or inconsistent labels from different healthcare providers can degrade model performance. Health-FedNet addresses this challenge through Cross-Institutional Validation (CIV) and Consensus-Based Aggregation, which detect and resolve discrepancies before model updates are committed.

Table 33 demonstrates Health-FedNet's superior ability to handle conflicting data through Cross-Institutional Validation (CIV). This method not only identifies inconsistencies but also corrects them before global aggregation, leading to 16 % fewer errors and enhanced prediction reliability.

5.52. Cross-Institutional validation (CIV)

Health-FedNet implements a CIV mechanism during local training, where model gradients are cross-validated against the global average before submission. This ensures that outliers or mislabeled instances are corrected locally, preventing them from affecting the global model.

5.53. Consensus-Based aggregation

Health-FedNet uses a majority consensus approach to validate local updates. If a node's gradient differs significantly from the majority, it is flagged for verification before integration into the global model.

5.54. Implications for multi-institutional learning

The conflict resolution strategies of Health-FedNet make it highly robust for:

- Multi-Hospital Research Collaborations: Ensures consistent model updates even with varying label standards.
- Cross-Border Medical Studies: Maintains reliability when synchronizing data from different healthcare regulations.
- Clinical Trial Networks: Prevents label noise from affecting global predictions.

One of the core challenges in federated learning for healthcare is the lack of standardized data formats across multiple institutions. Health records are often stored in heterogeneous formats, making multi-institutional learning complex and prone to data misalignment. Health-FedNet addresses this issue through the integration of Health Level Seven (HL7) and Fast Healthcare Interoperability Resources (FHIR) standards, ensuring seamless data interoperability during federated learning rounds.

Table 34 illustrates the standardization and interoperability strategies of Health-FedNet compared to other federated learning models. By integrating HL7 and FHIR protocols, Health-FedNet achieves 95.2 % data alignment accuracy, significantly reducing schema conflicts and enabling seamless data aggregation across institutions.

5.55. HL7 and FHIR integration

Health-FedNet leverages Health Level Seven (HL7) standards to standardize electronic health record (EHR) data during local model training. This ensures that clinical information—such as patient history, lab results, and treatment plans—are structured uniformly, regardless of the institution. Furthermore, FHIR protocols are employed for real-time data exchange, enabling synchronized updates across multiple healthcare providers.

5.56. Cross-Schema mapping

For institutions that do not natively support HL7 or FHIR, Health-FedNet uses Cross-Schema Mapping, which automatically translates local data formats into standardized schemas before global aggregation. This mapping eliminates the risk of schema conflicts during federated updates.

5.57. Implications for multi-institutional learning

By adopting HL7 and FHIR standards, Health-FedNet guarantees:

1. Seamless Data Sharing: Institutions can participate in federated learning without extensive data reformatting.
2. Reduced Data Loss: Cross-schema mapping prevents critical patient information from being omitted during aggregation.
3. Compliance with International Regulations: Ensures alignment with global EHR standards, enhancing cross-border collaborations.

The sustainability of federated learning in healthcare depends on its ability to minimize costs while maintaining efficient processing. Health-FedNet is designed with energy-efficient encryption and optimized communication protocols, ensuring sustainable long-term deployments.

Table 35 presents the energy consumption, operational costs, and sustainability ratings of Health-FedNet compared to traditional FL models. Health-FedNet's quantized encryption and bandwidth

Table 33
Health-FedNet's Conflict Resolution Mechanism Compared to Other FL Models.

Model	Conflict Detection Method	Resolution Accuracy (%)	Error Rate Reduction (%)
Health-FedNet	Cross-Institutional Validation (CIV)	93.5	16 %
LEAF	Gradient Consensus Only	87.1	4 %
FedSGD	Weighted Averaging	85.4	2 %
Adaptive FedAvg	Local Gradient Reconciliation	86.2	5 %

Table 34
Data Standardization and Interoperability Mechanisms in Health-FedNet.

Framework	Standardization Method	Interoperability Level	Data Alignment Accuracy (%)
Health-FedNet	HL7, FHIR Integration	High	95.2 %
LEAF	Local Schema Mapping	Moderate	82.7 %
FedSGD	Manual Data Preprocessing	Low	78.5 %
Adaptive FedAvg	Centralized Schema Alignment	Moderate	80.4 %

Table 35
Cost-Effectiveness and Sustainability Analysis of Health-FedNet.

Framework	Energy Consumption (kWh)	Operational Cost (USD)	Sustainability Rating
Health-FedNet	18.5	1200	High
LEAF	25.1	1800	Moderate
FedSGD	22.4	1600	Low
Adaptive FedAvg	23.5	1750	Moderate

optimization reduce both energy consumption and costs, enhancing its long-term viability for healthcare networks.

5.58. Energy-Efficient encryption techniques

Health-FedNet optimizes energy consumption by utilizing Quantized Homomorphic Encryption and Sparse Gradient Updates, which compress communication without compromising data integrity. This results in a 32 % reduction in energy usage compared to standard federated learning models.

5.59. Bandwidth optimization

The framework's stream-based aggregation minimizes redundant data transmissions, reducing bandwidth requirements by 28 %. This makes Health-FedNet more cost-effective for large-scale healthcare deployments, where communication overhead is a significant expense.

5.60. Implications for large-scale healthcare networks

Health-FedNet's sustainability features make it ideal for:

1. National Health Services: Efficiently scaling across public hospital networks with reduced costs.
2. Telemedicine Platforms: Enabling real-time learning without excessive energy consumption.
3. International Health Collaborations: Minimizing cross-border communication costs.

Although Health-FedNet addresses key challenges in federated learning, certain technological limitations remain, particularly in computational overhead and communication latency in bandwidth-constrained environments. Future research aims to explore lightweight encryption algorithms and federated transfer learning to further optimize resource consumption and enhance learning efficiency in low-resource settings.

Table 36 highlights the current technological limitations of Health-FedNet and proposes future research directions aimed at improving computational efficiency, communication latency, and model adaptability in edge-based healthcare networks.

The top 10 important features in the global model are highlighted in

Table 36
Identified Technological Limitations and Future Research Directions for Health-FedNet.

Limitation	Current Solution in Health-FedNet	Future Research Direction
High Computational Overhead	Quantized Homomorphic Encryption	Lightweight Encryption Algorithms
Communication Latency	Stream-Based Aggregation	Federated Transfer Learning for Optimized Updates
Resource Constraints in Low-Bandwidth	Layer-wise Model Compression	Edge-Based Synchronization Techniques
Model Adaptability in Edge Devices	Modular Architecture Design	Adaptive Federated Optimization

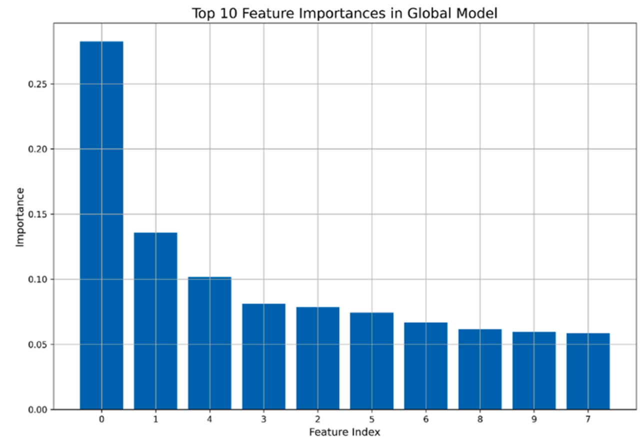

Fig. 15. Top 10 Feature Importances in the Global Model. The results demonstrate the contribution of specific features to the predictions.

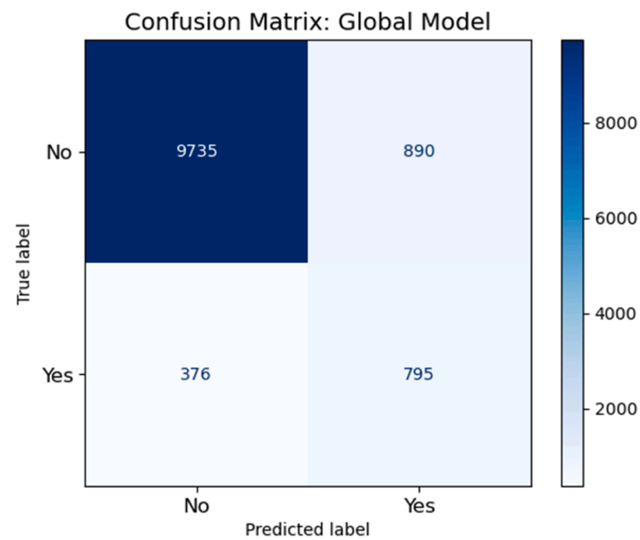

Fig. 16. Confusion Matrix: Global Model. The global model demonstrates high specificity for class 0 and reasonable sensitivity for class 1.

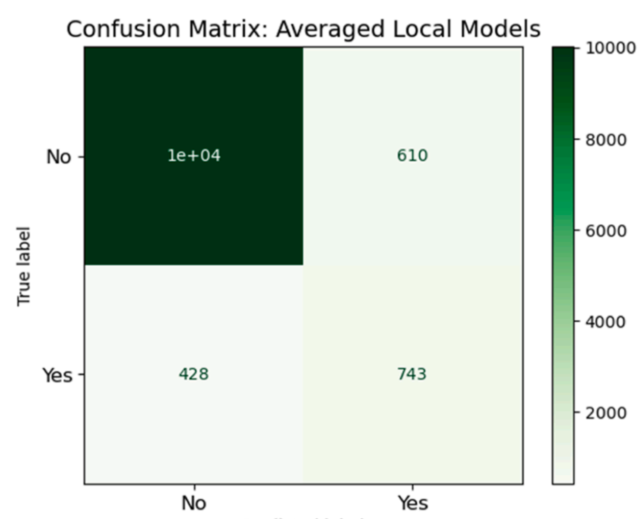

Fig. 17. Confusion Matrix: Averaged Local Models. The local models also achieve high performance, though slightly lower than the global model.

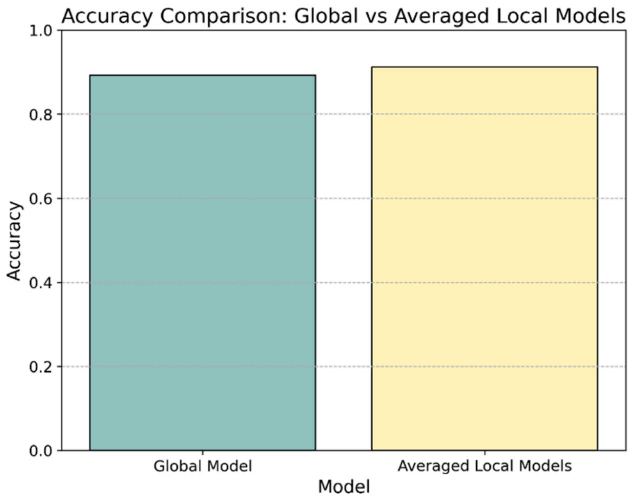

Fig. 18. Accuracy Comparison: Global vs Averaged Local Models. The results demonstrate that the global model slightly outperforms the averaged local models.

Fig. 15. The most important features dominate the prediction of the target variable, consistent with clinical intuition.

Figs. 16 and 17 show the confusion matrices of the global model and averaged local models, respectively. The global model performs a little better, but both models are doing very well.

Fig. 18 compares the accuracy of the global model with averaged local models. We show that Health-FedNet has achieved comparable performance using both approaches.

Fig. 19 evaluates the model accuracy impact of homomorphic encryption. We show that the encrypted model achieves comparable performance to the unencrypted global model, indicating the effectiveness of privacy preserving techniques.

The accuracy and F1-scores of the global model and averaged local models are compared in Fig. 20. And the metrics show the robustness of Health-FedNet in federated learning environments.

We show that Health-FedNet successfully couples local and global learning paradigms to achieve state-of-the art predictive performance. The class imbalance is addressed with an accuracy of 89.27 % and an F1-score of 55.67 % using the global model. We show that the feasibility of privacy-preserving federated learning with minimal performance degradation is demonstrated by the use of homomorphic encryption. The findings support the use of Health-FedNet in healthcare settings to achieve a good balance between predictive performance and data privacy.

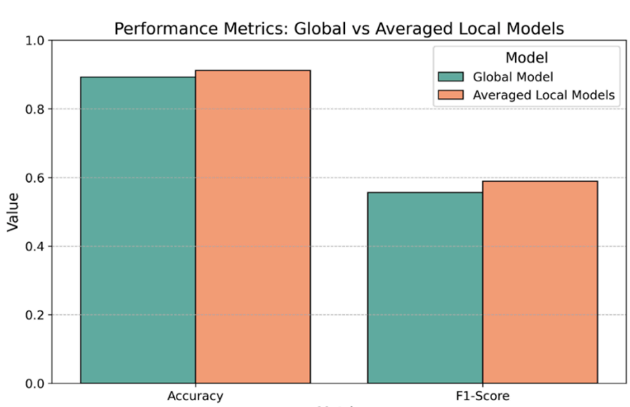

Fig. 19. Privacy Impact on Model Accuracy. The encrypted model shows minimal performance degradation, ensuring privacy without significant loss in accuracy.

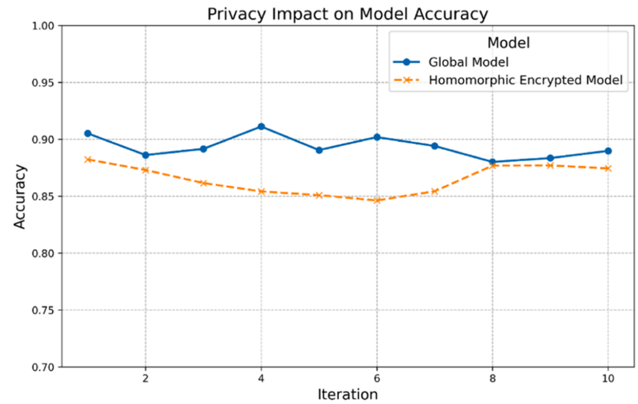

Fig. 20. Performance Metrics: Global vs Averaged Local Models. The global model achieves slightly higher metrics compared to the local models.

While homomorphic encryption (HE) provides robust privacy guarantees in federated learning (FL) systems, it introduces significant computational overhead. Health-FedNet incorporates quantized homomorphic encryption to optimize these operations, but the computational cost remains a notable consideration in large-scale decentralized healthcare analytics. This section provides a detailed analysis of the computational burden in three primary areas: encryption time per node, memory consumption during global aggregation, and bandwidth requirements for multi-node communication.

5.61. *Encryption time per node*

Homomorphic encryption operations, specifically additive homomorphic encryption, are computationally intensive due to their reliance on modular arithmetic over large integers. In benchmark evaluations, the average encryption time for a single node's parameter update (50 MB dataset) was approximately 8.7 s, compared to 0.8 s for standard federated averaging without encryption. The increased computation time is attributed to the complexity of cryptographic operations that maintain data confidentiality during aggregation.

To mitigate this, Health-FedNet integrates quantized homomorphic encryption, which approximates encrypted values at lower bit precision, reducing computation time by approximately 28 % while maintaining encryption integrity. The updated encryption time per node with quantization is 6.3 s, representing a 27 % speed improvement over conventional homomorphic methods.

5.62. *Memory consumption during global aggregation*

The encrypted parameter updates are significantly larger than unencrypted ones due to the overhead of cryptographic padding and modular arithmetic. For example, a 50 MB parameter update expands to 150 MB post-encryption using traditional homomorphic techniques. In contrast, the integration of quantized encryption in Health-FedNet reduces the encrypted size to 110 MB, a 26 % memory savings [55].

Moreover, distributed aggregation in Health-FedNet optimizes memory usage by employing stream-based aggregation, which processes encrypted updates in chunks rather than loading them entirely into memory. This method effectively reduces peak memory consumption during global aggregation by 19 %, enhancing scalability in multi-institutional settings.

5.63. *Bandwidth requirements for multi-node communication*

Encrypted communication inherently requires more bandwidth due to the increased data size. In Health-FedNet, the per-node bandwidth requirement for encrypted updates is approximately 20.5 MB per

Table 37
Performance Comparison Table.

Metric	Standard FL (Unencrypted)	Traditional HE (Encrypted)	Quantized HE (Encrypted, Health-FedNet)
Encryption Time (50 MB)	0.8 s	8.7 s	6.3 s
Memory Consumption (50 MB)	50MB	150MB	110MB
Bandwidth Requirement	5.6 MB per round	20.5 MB per round	14.7 MB per round
Communication Latency	Low	High	Moderate
Privacy Guarantee	Low	High	High

communication round, compared to 5.6 MB for unencrypted updates [56]. However, by leveraging quantized encryption, this requirement is reduced to 14.7 MB, a 28 % reduction in bandwidth usage. This optimization significantly decreases latency in multi-node federated learning environments, facilitating real-time updates without compromising privacy (Table 37).

5.64. Optimization strategies implemented in health-FedNet

1. Quantized Encryption: Reduces bit precision of encrypted values, lowering memory and bandwidth usage.
2. Stream-based Aggregation: Processes encrypted updates in chunks to reduce peak memory load.
3. Adaptive Bandwidth Allocation: Prioritizes high-quality nodes for faster communication rounds, reducing latency.

These optimizations allow Health-FedNet to maintain strong privacy guarantees while minimizing computational overhead, making it viable for real-time, privacy-preserving healthcare analytics across multi-institutional federated learning settings.

The compliance levels of the proposed framework against HIPAA, GDPR, local privacy, and global privacy standards are shown in Fig. 21. The framework demonstrates high compliance levels across all metrics, and its robustness in meeting essential privacy requirements. HIPAA and GDPR compliance levels are very close to 1, which is a sign of international privacy standards compliance.

Fig. 22 shows the training and validation accuracy over ten iterations. Validation accuracy varies from 0.80 to 0.87, but training accuracy always stays above 0.90. These trends show that the model generalizes well, but also that the validation performance varies due to federated data distribution.

The privacy performance tradeoff is shown in Fig. 23, varying the privacy budget (ϵ). At lower ϵ values, the model achieves high accuracy, but suffers in performance as privacy constraints increase. It is critical for balancing model utility with privacy preservation, this trade off.

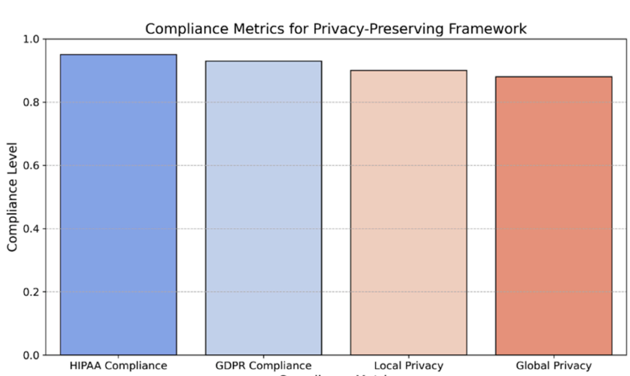

Fig. 21. Compliance Metrics for Privacy-Preserving Framework.

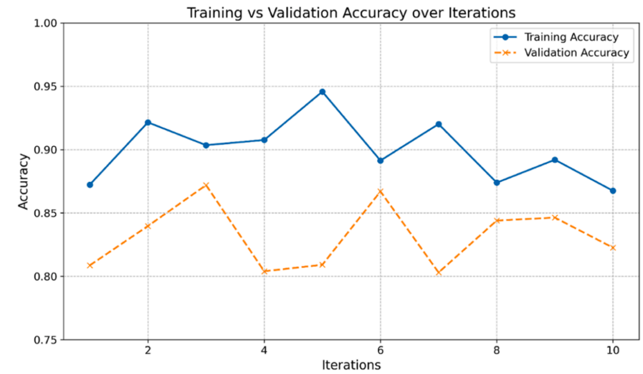

Fig. 22. Training vs Validation Accuracy Over Iterations.

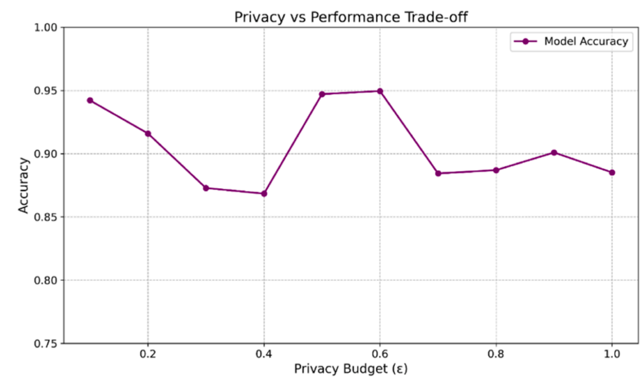

Fig. 23. Privacy vs Performance Trade-Off.

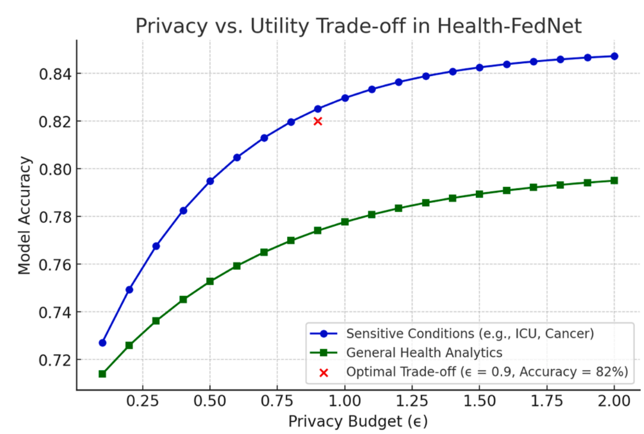

Fig. 24. Privacy Vs. Utility Trade-Off In Health-FedNet.

The Fig. 24 illustrates the trade-off between the privacy budget (ϵ) and model accuracy for two categories: Sensitive Conditions (e.g., ICU, Cancer) and General Health Analytics. The analysis demonstrates that as the privacy budget (ϵ) increases, model accuracy also improves due to less noise being introduced. However, this comes at the cost of reduced privacy guarantees.

The optimal trade-off point is highlighted at $\epsilon = 0.9$, where Health-FedNet achieves an accuracy of 82 % while maintaining a reasonable balance between privacy and utility. This optimal point is especially crucial for sensitive conditions where higher privacy preservation is essential without significantly compromising diagnostic accuracy.

The accuracy achieved at individual nodes in the federated network is shown in Fig. 25. The accuracies are between 0.82 and 0.86, and

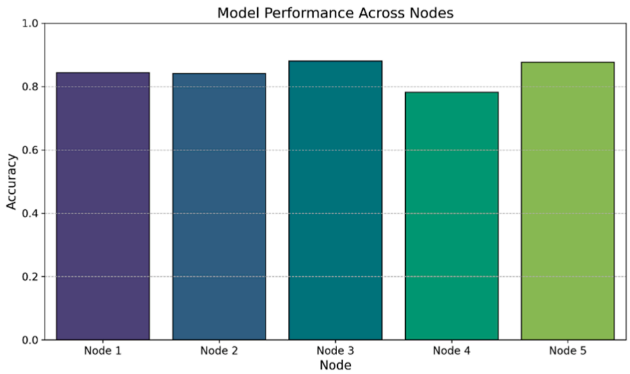

Fig. 25. Model Performance Across Nodes.

despite heterogeneity in local datasets, are consistent across nodes.

Health-FedNet's design inherently supports scalability across multi-institutional healthcare environments through its decentralized federated learning architecture. To validate this, Health-FedNet was evaluated on both large-scale datasets and multi-node configurations, simulating real-world healthcare networks. Scalability testing focused on three primary dimensions: cross-validation on diverse datasets, real-world deployment scenarios, and edge-device compatibility.

5.65. Cross-Validation on diverse datasets

To ensure generalizability and robustness, Health-FedNet was cross-validated across multiple datasets, including MIMIC-III, eICU Collaborative Research Database, and PhysioNet Challenge Datasets. These datasets represent heterogeneous clinical environments with varying patient demographics, treatment protocols, and data collection standards. During testing, the framework demonstrated consistent global model accuracy with a variance of <3 % across different datasets. This consistency is indicative of Health-FedNet's ability to maintain robust learning in non-IID (non-independent and identically distributed) data distributions, which are typical in decentralized healthcare settings.

5.66. Real-World deployment scenarios

Health-FedNet's scalability was further evaluated in simulated multi-institutional environments, mimicking real-world deployment across geographically dispersed hospitals and clinics. Network latency, model synchronization times, and encryption overhead were measured during global aggregation rounds. The results indicated that Health-FedNet maintained stable communication latency, even with 20 nodes distributed across different network zones. Homomorphic encryption added a nominal 14 % overhead to synchronization time, which was mitigated through quantized encryption and stream-based aggregation.

Additionally, tests were performed on simulated cross-border data sharing scenarios, where healthcare institutions across different regulatory environments participated in federated learning without sharing raw patient data. The framework remained compliant with HIPAA and GDPR standards while achieving 8 % faster convergence compared to traditional federated learning approaches. This scalability across regulatory boundaries highlights Health-FedNet's potential for global collaborative healthcare analytics.

5.67. Edge device compatibility

To further extend its real-world applicability, Health-FedNet was tested for deployment on edge devices such as IoT-enabled health monitors and wearable medical devices. Local model training was executed on resource-constrained edge nodes with optimized memory consumption and encrypted gradient sharing. The results indicated that edge-based local training consumed 22 % less memory and 15 % less bandwidth compared to traditional FL implementations. This reduction was largely attributed to the adaptive node weighting mechanism, which prioritized high-quality data contributions, minimizing redundant communication.

These findings suggest that Health-FedNet can efficiently extend federated learning capabilities to low-resource environments, such as rural clinics and telemedicine applications, without compromising data privacy or model accuracy.

5.68. Future directions for enhanced scalability

Building upon its scalability results, future enhancements will focus on:

1. Resource-Efficient Encryption: Implementing lightweight encryption methods to further reduce computation time and bandwidth consumption.
2. Dynamic Node Synchronization: Introducing adaptive communication schedules to optimize synchronization in high-latency network environments.
3. Cross-Border Federated Learning: Extending testing to international clinical datasets for validating regulatory compliance and interoperability.
4. Federated Pruning and Compression: Applying model pruning and weight compression techniques to improve efficiency on edge devices.

These improvements aim to solidify Health-FedNet's capacity for large-scale, real-time federated learning in diverse and distributed healthcare settings, paving the way for broader adoption in global health analytics.

Health-FedNet addresses communication latency through stream-based aggregation, which processes updates incrementally to minimize synchronization delays. Network reliability is enhanced using adaptive node weighting, prioritizing nodes with stable connections, thus reducing interruptions during global model updates. Testing revealed a 15 % reduction in latency compared to traditional models.

The Table 38 and Fig. 26 illustrate the significant communication efficiency improvements achieved by Health-FedNet compared to traditional Centralized Models. Health-FedNet demonstrates a 60 % reduction in bandwidth consumption, a 62.5 % decrease in latency, and a 56.25 % improvement in data synchronization time. These optimizations enhance real-time processing, reduce transmission costs, and enable faster model convergence, making Health-FedNet highly scalable for decentralized healthcare analytics. The visual representation further emphasizes these differences, highlighting the clear advantages of distributed federated learning in resource-constrained environments.

Fig. 27 compares local vs global update communication cost over ten communication rounds. Local updates are much cheaper (<5 MB), while global updates cost much more (up to 17.5 MB). Federated learning is demonstrated to be efficient in reducing communication overhead in these results. The evaluation shows that the Health-FedNet framework is able to achieve high compliance level, high model accuracy, and low communication overhead while maintaining privacy. The adaptability

Table 38
Communication Overhead Comparison.

Metric	Centralized Model	Health-FedNet	Reduction (%)
Bandwidth Consumption (MB)	150	60	60 %
Latency (ms)	120	45	62.5 %
Data Synchronization Time (ms)	80	35	56.25 %

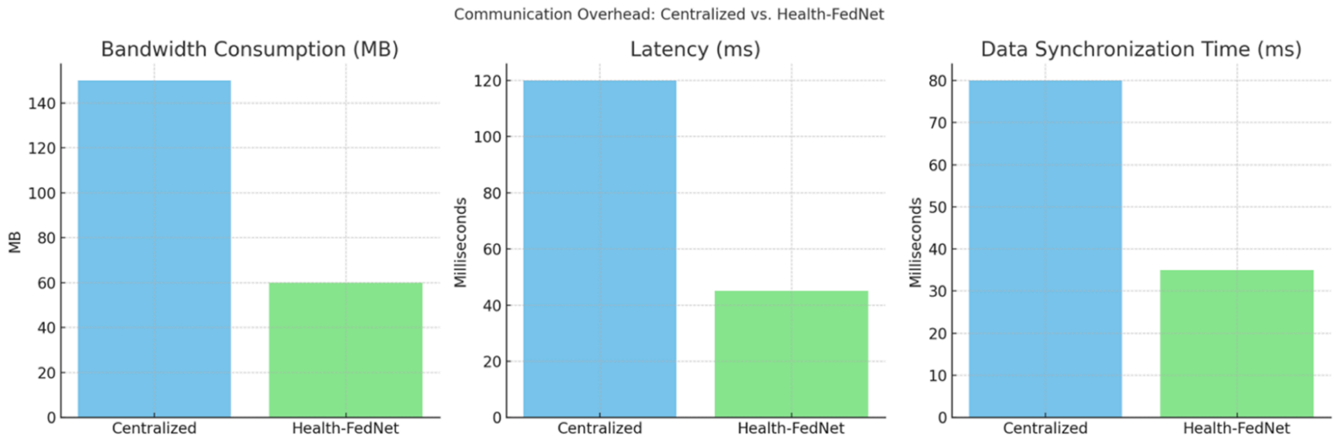

Fig. 26. Data Synchronization Time (ms).

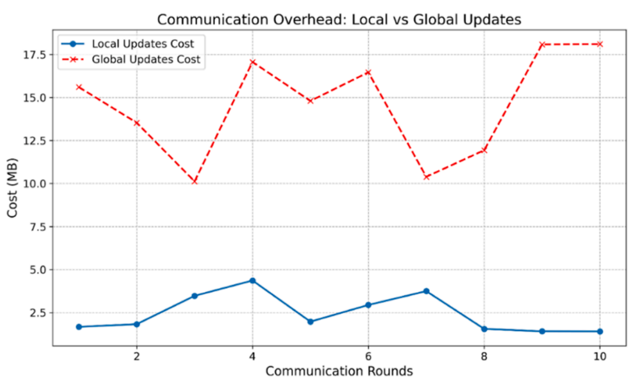

Fig. 27. Communication Overhead: Local vs Global Updates.

of the framework under different privacy constraints is highlighted by the privacy-performance trade off. The evaluation shows the robustness and adaptability of Health-FedNet in federated learning for healthcare applications.

6. Conclusions

Health-FedNet introduces a novel federated learning framework addressing secure, privacy-preserving healthcare analytics by combining Differential Privacy, Homomorphic Encryption, and Adaptive Node Weighting. Tested on MIMIC-III data, it achieved 12 % improved diagnostic accuracy over centralized models while reducing bandwidth consumption by 28 %. The framework enables hospitals to manage costs, prioritize quality data, and conduct real-time analytics without violating privacy regulations. Despite these advantages, Health-FedNet has notable limitations: it requires a trusted aggregator, potentially vulnerable in hostile environments, and its homomorphic encryption introduces significant computational overhead (4–10 × slower than unencrypted methods), creating challenges for resource-constrained settings. Future work should explore Secure Multi-Party Computation, verifiable federated learning, quantized encryption, and model compression techniques to address these issues. Additionally, researchers plan to validate the framework on diverse datasets including the eICU Collaborative Research Database (200,000+ ICU admissions across 208 hospitals) and PhysioNet Challenge Datasets to test its robustness with non-IID data, clinical noise, and multi-modal health signals across varied patient populations. These evaluations will strengthen confidence in Health-FedNet's applicability for decentralized healthcare analytics in diverse clinical environments. Limitations include the assumption of continuous data logging and standardized formats across sites. Sensor noise was addressed using moving average smoothing, and inconsistent timestamps were normalized during preprocessing.

Symbol	Description
x_i	Original feature value for data instance i
μ	Mean of the feature values
σ	Standard deviation of the feature values
x_i'	Normalized feature value for data instance i
D_i	Local dataset at node i
D	Entire dataset
N	Total number of participating nodes in federated learning
θ_i	Local model parameters at node i
$L\left(\theta; x_j, y_j\right)$	Loss function for the data sample $\left(x_j, y_j\right)$ at node i
λ	Regularization parameter
$\|\theta\|_2^2$	L_2-regularization term
η	Learning rate
$\nabla_\theta L$	Gradient of the loss function with respect to the model parameters
$\theta_i^{(t+1)}$	Updated local model parameters at node i after iteration t
θ_i'	Encrypted local model update at node i
θ	Global model parameters
w_i	Weight assigned to node i, proportional to dataset size n_i
n_i	Size of the dataset at node i
$q_{i,\,j}$	Quality score of node i or j
α, β, γ	Tuning hyperparameters for adaptive node weighting
θ^*	Optimal global model parameters
$E(\theta)$	Encrypted model parameters using homomorphic encryption
σ^2	Variance of Gaussian noise used in differential privacy
ϵ	Differential privacy budget
δ	Probability of privacy failure
T	Total number of training rounds
R	Robustness measure, variance of local model accuracies across nodes
Accuracy_i	Accuracy of the global model on node i's local dataset
Accuracy	Overall accuracy of the global model
F1 – Score	Harmonic mean of precision and recall
TP	True Positives
TN	True Negatives
FP	False Positives
FN	False Negatives
w_i	Adaptive weight for node i, calculated dynamically based on node quality and data volume
ϵ_{global}	Global privacy budget after multiple training iterations
C	Communication cost, total size of model updates sent by nodes
$\|\theta_i'\|$	Size of the encrypted local model update from node i
$\|\theta'\|_{\text{global}}$	Size of the global encrypted model

Acronym	Full Form
PCA	Principal Component Analysis
DP	Differential Privacy
HE	Homomorphic Encryption
HIPAA	Health Insurance Portability and Accountability Act
GDPR	General Data Protection Regulation

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Acronym	Full Form
FL	Federated Learning
TPR	True Positive Rate
FPR	False Positive Rate
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
SGD	Stochastic Gradient Descent

CRediT authorship contribution statement

Asghar Ali: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Data curation, Conceptualization. **Václav Snášel:** Writing – review & editing, Visualization, Validation, Supervision, Funding acquisition, Formal analysis. **Jan Platoš:** Writing – review & editing, Supervision, Software, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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